

Experimental study on the combined compression-bending-torsion behavior of prestressed concrete towers for wind turbines considering horizontal joints

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ABSTRACT

Prestressed concrete towers for wind turbines are generally subjected to combined compression-bending-torsion loads due to prestressing force and aerodynamic force. The tower segments and foundation are connected through horizontal joints, whose connection performance mainly relies on the external post-tensioned prestressing strands. Consequently, the prestressing force may not be able to compact the entire joint under extreme loads, and the joint will open, resulting in complex nonlinear behavior of the tower under combined loads. This paper presents an experimental study on the combined behavior of prestressed concrete tower, considering effects of horizontal joints. Specimens with varying prestress ratios and bending-torsion ratios were tested under cyclic loading. The key indicators such as failure modes, ultimate capacity, hysteretic performance, and prestress variations were discussed, revealing the combined compression-bending-torsion mechanism of prestress concrete towers. The results demonstrate that increasing prestress ratios and bending-torsion ratios can enhance the ultimate capacity, while ductility improves with lower prestress ratios and higher bending-torsion ratios. Several calculation methods of bending and torsional capacity in existing design methods were evaluated based on test results. The ultimate capacities were underestimated by the methods controlled by horizontal joints but overestimated by the methods governed by tower segments. For towers with high safety requirements, the methods controlled by joints remain applicable as conservative yet expeditious preliminary tools for conceptual stages.

1. Introduction

With the fast development of wind turbines, hybrid towers (consisting of upper steel tower and lower prestressed concrete tower, as shown in Fig. 1) have gained widespread applications as replacements for traditional steel tubular towers owing to the superior load-bearing capacity and cost advantages, effectively contributing to cost reduction and efficiency improvement in wind energy utilization. The load-bearing capacity advantages of the hybrid tower are primarily provided by the lower prestressed concrete tower. To significantly reduce self-weight, prestressed concrete towers are typically designed as thin-walled structures with large hollow ratios. Considering transportation and lifting requirements, prestressed concrete towers are generally constructed as segmental assembled structures. Precast concrete components are first assembled into tower segments through vertical joints at the construction site and subsequently assembled into a complete

tower through horizontal joints [1] (Fig. 1). The vertical joints employ densely arranged connecting reinforcement and high-strength grouting materials cast on-site, with their connection strength typically exceeding that of the precast components themselves [2]. However, horizontal joints lack connecting reinforcement and rely solely on grouting materials or structural adhesives at the interface, making it challenging to ensure stable connections [3]. This necessitates the application of continuous post-tensioned prestress along the vertical direction to integrate various tower segments with the foundation. External unbonded post-tensioned prestress are most adopted, where steel strands are positioned within the hollow interior of the tower and anchored at both ends - to the foundation at the tower base and the adapter at the tower top. The compression induced by prestressing strands significantly strengthens the horizontal joint connections, thereby enhancing the lateral stiffness of the tower [4]. Nevertheless, the prestressing force may not be able to compact the entire joint under extreme loads, resulting in complex behavior of the tower, which is the

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Nomenclature			
m	Bending-torsion ratio	A_c	Cross-sectional area
B	Torsional arm length	A_s	Section area of steel reinforcement bars
n	Prestress ratio	A_{sv}	Section area of vertical reinforcement bars
P	Prestressing force	A_{sh}	Section area of horizontal reinforcement bars
σ_{p0}	Average initial prestress	A_p	Section area of prestressing steel strands
M	Bending moment	A_o	Area enclosed by the middle diameter
T	Torque	f_c	Axial compressive strength of concrete
N	Axial Force	$f_{c,cube}$	Cubic compressive strength of concrete
F	Lateral force	f_t	Axial tensile strength of concrete
H	Height of the tower segment	f_{sy}	Yielding strength of steel reinforcement bars
Δ	Lateral displacement	f_{py}	Yielding strength of prestressing steel strands
Δ_r	Residual displacement	M_{ue}	Measured ultimate bending moments
CI	Capacity index	M_{uc1}	Ultimate bending moment calculated by joint-controlled method
DI	Ductility index	M_{uc2}	Ultimate bending moment calculated by segment-controlled method
ε	Strain	T_{ue}	Measured ultimate torque
t	Thickness of tower wall	T_{uc1}	Ultimate torque calculated by joint-controlled method
u_o	Perimeter corresponding to the middle diameter	T_{uc2}	Ultimate torque calculated by segment-controlled method

key and difficult point in investigations of prestressed concrete towers.

Current research on prestressed concrete towers predominantly focused on investigating global structural responses such as stiffness, frequency, buckling, and vibration characteristics [5–8], while structural optimization analyses and designs were conducted based on strength, fatigue resistance, and dynamic responses [9–12]. These studies provide valuable references for understanding the overall structural behavior of prestressed concrete towers. However, most analyses of ultimate capacity have employed elastic stress verification methods based on material strength [13,14], failing to precisely evaluate the load-bearing performance of towers. In recent years, experimental investigations on prestressed concrete towers have been conducted. Li et al. [15] performed compression-bending tests on a 1:10

scaled hybrid tower model, yet their study neither considered joint effects nor implemented destructive loading, thereby precluding observation of ultimate behaviors. Ma et al. [16] conducted model tests on a 5MW octagonal prestressed concrete tower, developing a 6.6m-high prototype with a 1:15 scaling ratio to examine mechanical performance and dynamic characteristics. However, the specimen also omitted joint details and avoided destructive loading tests. Most existing studies assume non-opening horizontal joints, thereby neglecting the nonlinear effects of joint behavior. Drawing upon results in similar prefabricated structures such as precast bridge piers and tunnel segments [17–21], recent studies have gradually recognized the critical effect of horizontal joint. Zhang et al. [22] conducted large-scale (1:3) model tests with detailed consideration of horizontal joints, vertical joints, and door

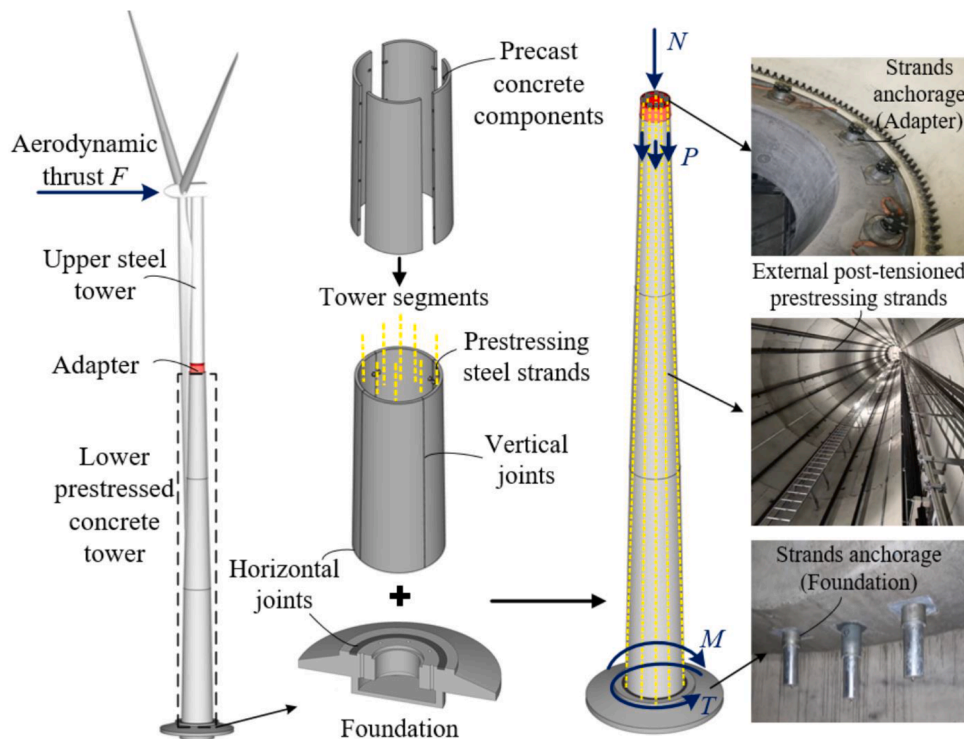


Fig. 1. Schematic of structures and loads of prestressed concrete towers.

openings, though the test was limited to elastic loading without analyzing ultimate mechanical performance. Tan [23,24] experimentally investigated the bending behavior of horizontal joints in prestressed concrete towers, revealing that the joints could open and close under bending moments while maintaining favorable self-centering capability. Wang [25] numerically simulated prestressed concrete towers with horizontal joints, demonstrating the bending moment can be still greatly increased after joint opening. The authors [4] developed a full-stages theoretical analysis model considering horizontal joint effects for towers under combined compression-bending loads based on experimental investigations. These findings collectively indicate that horizontal joint opening is significantly influenced by prestressing forces, and that joint behavior substantially affects the mechanical performance of tower.

However, existing research on prestressed concrete towers has predominantly focused on bending behavior, while the torque is also a critical load component acting on towers [26]. The primary loads are illustrated in Fig. 1. Aerodynamic wind load constitutes the most significant external force on towers, primarily manifested as aerodynamic thrust F generated by rotor rotation. This thrust F generates substantial bending moment M , particularly at the tower base. Furthermore, during operations of wind turbines such as pitch adjustment and yaw orientation, the eccentric action of aerodynamic thrust F induces essentially uniform torque T along the tower height. It should be noted that as prestressed concrete towers are typically located in the lower portion of hybrid towers, the application point of aerodynamic thrust F is considerably distant from the tower base. The high shear-span ratio of the tower results in negligible shear effects. After tensioning steel strands, the prestressing system applies substantial prestressing force P at the tower top. Concurrently, the combined self-weight of the wind turbine and tower generates vertical load N . These two components collectively form the compression in the prestressed concrete tower, though the vertical load N is typically significantly smaller than prestressing force P . Different wind turbine control strategies and prestressing force magnitudes will alter the proportion of bending moment, torque, and compression, maintaining the tower in a continuously variable combined compression-bending-torsion loading state. Therefore, the combined behavior should be considered in the investigation of prestressed concrete towers.

Numerous studies have investigated the combined compression-bending-torsion behavior of conventional structures such as building columns and bridge piers [27–31], systematically revealing critical mechanisms including bending-torsion failure modes and combined load-bearing capacities. Regarding wind turbine towers, investigations have been conducted on the combined behaviors of steel towers and concrete-filled steel tubular towers [32–35]. Studies have demonstrated that for slender thin-walled towers, torsional effects exhibit remarkable sensitivity to bending performance, and steel tubes are prone to bending-torsional buckling. However, as these structures lack horizontal joints and prestressing systems, the mechanical behavior differs from prestressed concrete towers. Limited research has addressed the combined bending-torsion performance of prestressed concrete towers. Haar and Marx [36] proposed calculation methods for bending capacity and shear-torsion resistance of horizontal joints, yet their approaches neglected the influence of horizontal joint opening. Kang et al. [37] numerically investigated the shear-torsion behavior after joint opening, revealing progressive capacity reduction with increasing opening displacements, and exposing limitations in Haar's method [36]. Chen et al. [38] developed torsional capacity formulas, but these studies presupposed torsional failure at horizontal joints while ignoring torsion effects of tower segments. Klein et al. [39] constructed a 1:10 scaled prestressed concrete tower model incorporating dry-connected horizontal joints. The non-destructive loading tests were conducted under combined compression-bending-shear-torsion loads and the strain distribution and deformation characteristics of the horizontal joints were analyzed. The results revealed non-uniform normal stress distributions

at horizontal joints, demonstrating inconsistencies between current elastic design assumptions and actual mechanical responses. Fülll et al. [40] conducted an experimental investigation to determine the torsional behavior of dry horizontal joints of prestressed concrete towers, analyzing the torsional capacity of horizontal joints under different bending moments. However, the tower segments remained intact throughout testing, leaving the coupling effects between tower segments and horizontal joints unexamined. It can be found that the research on the combined compression-bending-torsion behavior of prestressed concrete towers is still rare. In summary, destructive experimental investigations on prestressed concrete towers under combined compression-bending-torsion loading remain scarce to date. The coupling failure mechanisms between tower segments and joints under combined loading is not clear. No consensus has been established concerning the mechanical behavior of prestressed concrete towers subjected to combined compression-bending-torsion loads.

To address these gaps, this study conducts an experimental study on the combined compression-bending-torsion behavior of prestressed concrete tower considering effects of horizontal joints, based on previous research on similar structural systems. Through cyclic loading tests, systematic analyses were performed to evaluate the load-bearing capacity and hysteretic performance under combined loading. The test results elucidated the influence of critical parameters including bending-torsion ratio and prestressing force on structural behavior, while revealing the coupling failure mechanisms between tower segments and horizontal joints. Finally, the accuracy and applicability of bending and torsional capacity calculation formulas in existing design methods were evaluated based on the test results.

2. Experimental programme

2.1. Specimens design and manufacture

Given that the horizontal joints at the base typically govern the behavior of prestressed concrete towers, and vertical joints generally exhibit sufficient strength comparable to cast-in-place concrete, the experimental study focuses on the bottom tower segment and its horizontal joint with the foundation as simplified mechanical units. Eight specimens were designed for the cyclic loading test under combined compression-bending-torsion loads. Each specimen comprises a tower segment and a base block, connected through a horizontal joint as illustrated in Fig. 2.

The cross-sectional dimensions of the tower segment were scaled based on typical hollow ratios in engineering practice, with a diameter of 610 mm and wall thickness of 55 mm. Considering the small tapers adopted in most towers, which have negligible influence on the ultimate load-bearing performance, all specimens were designed as straight segments. In practical engineering, the high shear-span ratio of the tower results in negligible shear effects. The height of the tower will only affect the ratio of the bending moment to lateral force. Considering that the focus of this study is the combined load-bearing performance of the tower near the joint region, and it is impractical to achieve proportional height scaling under laboratory conditions, the specimen height was determined to eliminate shear effects of lateral forces while enabling bending-torsion loading. Following recommendations in [29], the shear-span ratio should be maintained above 4. Based on this, the total segment height of the specimen was set at 2600 mm, consisting of a 2400 mm standard segment and a 200 mm transition segment. The transition segment at the top features a solid cross-section to anchor prestressing strands and connect with test setup, simulating the adapters in hybrid towers. A double-layer bidirectional reinforcement layout with 4 mm diameter steel bars was adopted for the standard segments to form the reinforcement cage. The vertical reinforcement ratio is 1.00 % and the horizontal reinforcement ratio is 0.61 %, as detailed in Fig. 3.

A high-strength concrete base block (1200 mm diameter $\times$ 300 mm height) was designed to prevent damage of base. Eight 15.2 mm-

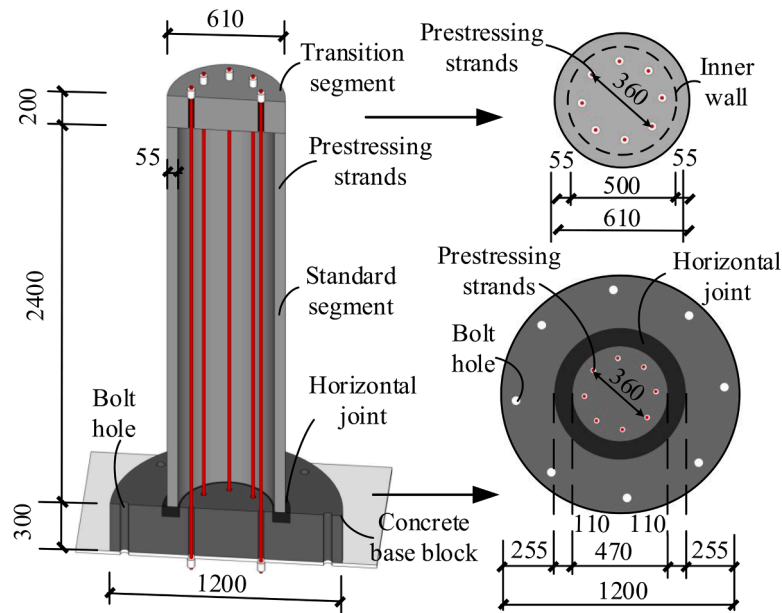


Fig. 2. Schematic of the specimen design.

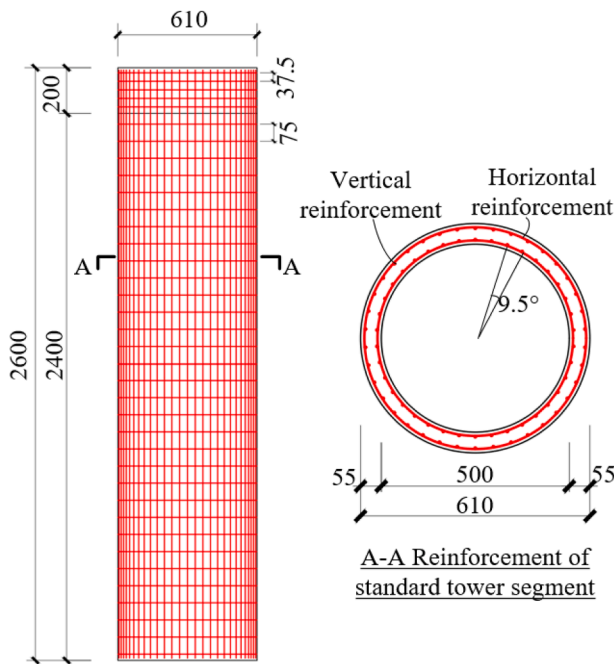


Fig. 3. Schematic of the reinforcement layout.

diameter 1860-grade prestressing strands were uniformly arranged along the circumferential direction. These strands were anchored at the top surface of the transition segment, passed through the interior hollow section of the tower segment, and anchored at bottom surface of the concrete base block, ensuring uniform prestress application and tight horizontal joint connection.

The reinforcement cage of the standard segment was integrally tied with the transition segment's reinforcement bars, followed by concrete casting to form a unified tower segment. Given the thin-walled configuration (55 mm wall thickness), fine-aggregate concrete with reduced maximum aggregate size was employed. During casting, a vibrator was operated at multiple circumferential positions to ensure concrete compaction. A groove was fabricated at the top surface of the concrete

base block. High-strength grout was poured into this groove to form the horizontal joint between the base and tower segment. Prior to grouting, three leveling blocks made of identical grout material were cast within the groove (Fig. 4). The tower segment was then hoisted and positioned atop these blocks, followed by groove grouting. The horizontal joint was finalized after natural curing. Subsequently, the prestressing strands were tensioned on the top of the steel cover using a sequential symmetric procedure. This staged tensioning protocol was implemented to ensure uniform prestress distribution and eliminate eccentric compression during strand activation. Material properties of the concrete, steel reinforcement, and strand were obtained through material tests based on material testing codes [41,42], and the results are summarized in Table 1.

The test parameters include the prestress ratio (n) and bending-torsion ratio (m). The prestress ratio was defined as the axial compression ratio induced by prestressing force ($n = P / f_c \cdot A_c$), with designed values ranging from 0.11 to 0.44 to investigate the effects of prestress magnitude. The bending-torsion ratio represents the proportion between bending moment and torque generated by lateral force. Based on typical loading characteristics of practical prestressed concrete towers and constraints of experimental condition, two bending-torsion ratios ($m = 4.125$ and 8.25) were selected, corresponding to torsional arm lengths of 800 mm and 400 mm respectively. The detailed parameters of all specimens are summarized in Table 2.

2.2. Test setup

A composite test setup was developed to apply combined compression-bending-torsion loads, comprising five primary components: rigid girder, bidirectional joint, servo actuator, steel cover, and steel base (Fig. 5). The bottom of the specimen was rigidly connected to the steel base using M46 high-strength bolt rods, while the top was secured to the steel cover. Prestressing strands were also anchored at the steel cover to apply prestress, replicating the connection methodology between adapters and upper steel tower sections in the hybrid towers. The steel cover's top surface was bolted to the rigid girder using high-strength bolts. The servo actuator, fixed to the reaction wall at one end, interfaced with the rigid girder through the bidirectional joint.

Offset from the specimen's central axis, the actuator generated bending moment (M) and torque (T) through the lateral force, superimposed with prestressing force (P) to achieve combined compression-

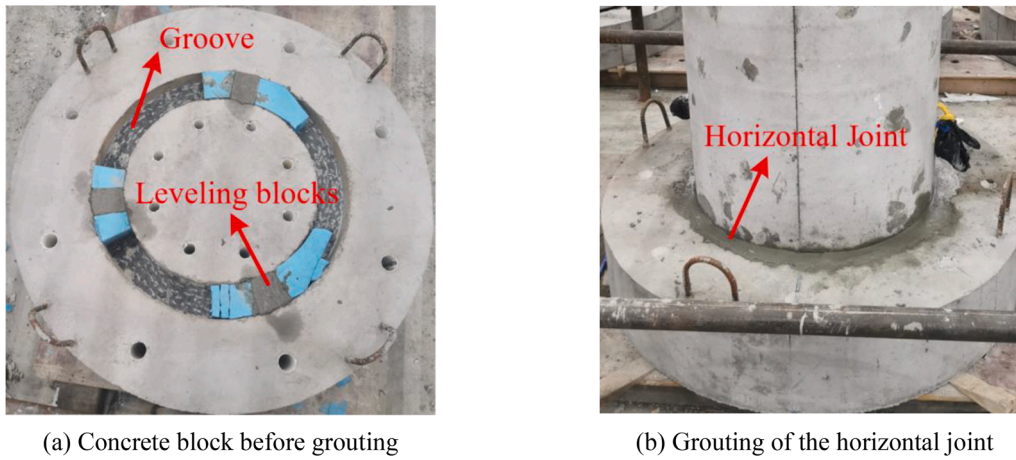


Fig. 4. Construction of the horizontal joint.

Table 1

Material properties of concrete, grout and steel.

Material type	Concrete of segment	Concrete of base block	Grout	Reinforcement bars	Steel strands
Compressive strength $f_{c,cube}$ /Yielding strength f_y (Mpa)	47.27	69.81	83.52	523.18	1804.86
Ultimate strength f_u (Mpa)	\	\	\	580.65	1996.43
Elastic modulus (Mpa)	32,687.67	37,528.25	39,767.3	199,325.2	204,021.3

Table 2

Test parameters of specimens.

Specimen label	Bending-torsion ratio m	Torsional arm length B (mm)	Designed prestress ratio n	Practical prestressing force P (kN)	Average prestress σ_{p0} (MPa)
CBT-800-0.11	4.125	800	0.11	482.7	431
CBT-800-0.22	4.125	800	0.22	854.1	762.6
CBT-800-0.33	4.125	800	0.33	1273.7	1137.2
CBT-800-0.44	4.125	800	0.44	1588.6	1418.4
CBT-400-0.11	8.25	400	0.11	470.8	420.3
CBT-400-0.22	8.25	400	0.22	864.9	772.2
CBT-400-0.33	8.25	400	0.33	1252.7	1118.5
CBT-400-0.44	8.25	400	0.44	1571.0	1402.7

Note: The first number in the specimen label represents the torsional arm length B ; the second number represents the prestress ratio n .

bending-torsion loading. Torsional arm length adjustment was achieved by altering the actuator's connection position relative to the centerline of rigid girder (400 mm or 800 mm offsets, as show in Fig. 6). This configuration enabled adjustment of bending-torsion ratios. It can be found that the torsional moment component decreases proportionally as the actuator is positioned closer to the centerline of rigid girder.

Upon application of lateral force, the specimen exhibits combined bending and torsional deformations, with the deformation relationship schematically illustrated in Fig. 7 [28,32]. Based on the geometric configuration, the torsional rotation angle (θ) and displacement angle (φ) can be calculated as:

$$\theta = \arcsin[(\Delta - \Delta_c)/B] \quad (1)$$

$$\varphi = \Delta_c \cdot H \quad (2)$$

where Δ is the horizontal displacement at the loading point and Δ_c is the horizontal displacement at the top of the specimen.

The bending moment (M) and the torque (T) acting on the bottom cross-section can be determined by:

$$M = F \cdot H_L \quad (3)$$

$$T = F \cdot B \sin \beta \quad (4)$$

where H_L denotes the vertical distance from the loading point to the

bottom of the tower segment.

Combining the above equations, the bending-torsion ratio (m) can be obtained as:

$$m = \frac{M}{T} = \frac{FH_L}{FB \sin \beta} = \frac{H_L}{B \sin \beta} \quad (5)$$

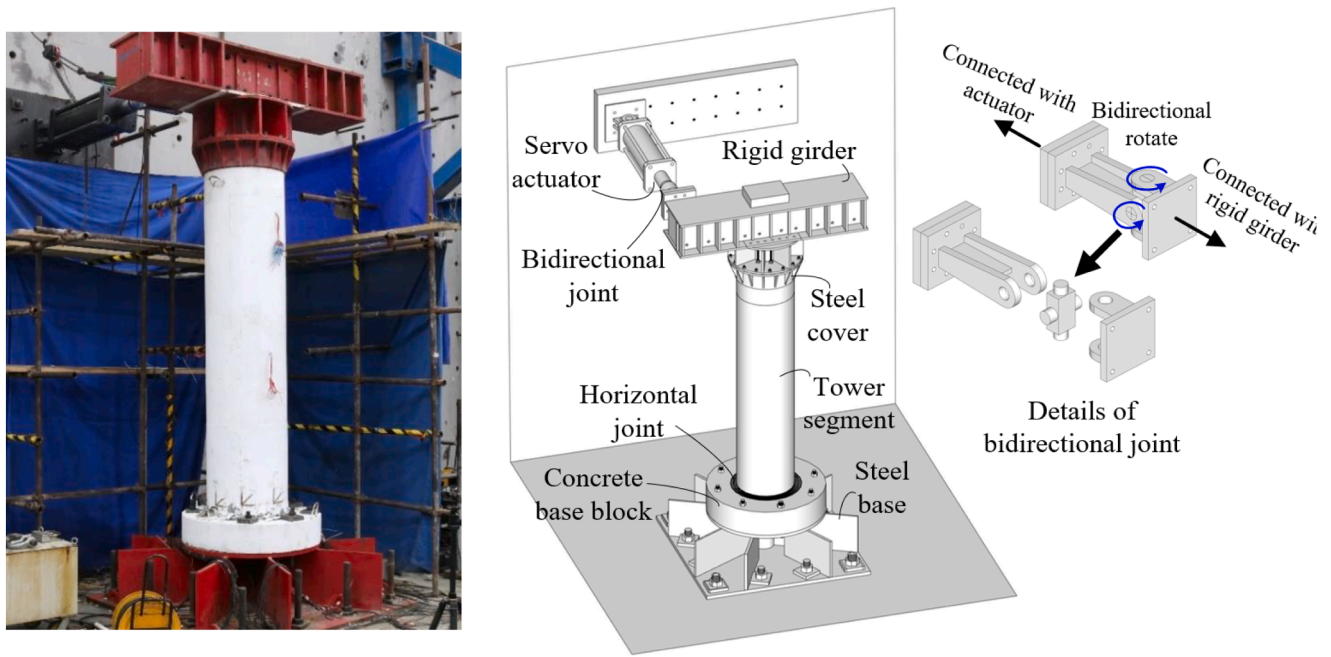
Given the minor torsional rotation ($\sin \beta \approx 1$), the ratio can be simplified to:

$$m = \frac{H_L}{B} \quad (6)$$

As H_L is constant, varying the torsional arm length (B) enabled controlled adjustment of bending-torsion ratios. The larger the torsional arm length, the lower the bending-torsion ratio. The values of m listed in Table 2 were obtained through this formula.

2.3. Measurements and loading scheme

The measurements include lateral force (F) from the servo actuator, horizontal displacement at the loading point (Δ), horizontal displacement at the top of the specimen (Δ_c), prestressing forces of strands (P_1 - P_8), and strains of reinforcements (ϵ). The lateral force was monitored by a load cell, while displacements were recorded using high-precision linear variable differential transformers (LVDTs), with sensor layout detailed in Fig. 8(a)(c). The prestressing forces were recorded by eight



(a) Photo during testing

(b) Schematic view

Fig. 5. Test setup.

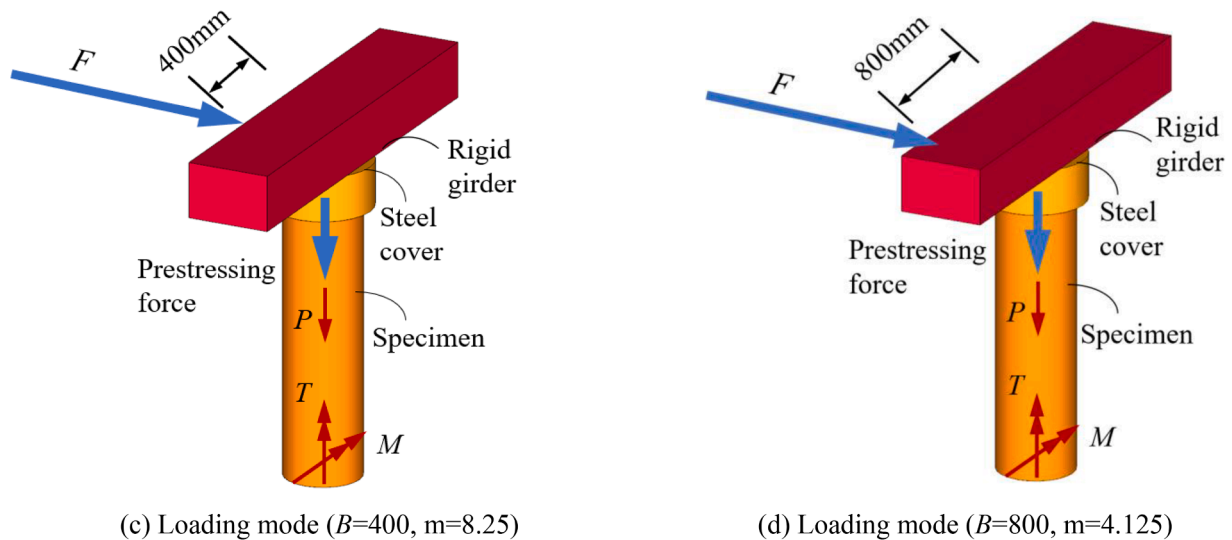
(c) Loading mode ($B=400$, $m=8.25$)(d) Loading mode ($B=800$, $m=4.125$)

Fig. 6. Schematic of loading modes.

load cells arranged at the bottom of the concrete base, as can be seen in Fig. 8(b).

Strain gauges were installed on the tower segment's bottom reinforcement, arranged at 45° circumferential intervals. This configuration comprised eight vertical strain gauges on vertical rebars (V_1 – V_8) and eight horizontal strain gauges on horizontal rebars (H_1 – H_8), as illustrated in Fig. 9.

Displacement-controlled cyclic loading was applied with displacement increments of ± 3 mm per loading level. This study focuses on the compression-bending-torsion capacity under cyclic loading rather than seismic performance such as cumulative damage and strength degradation under earthquake. Therefore, each level only included one loading cycle for simplification. Each target displacement of loading level was maintained for a 2-minute holding period to facilitate data acquisition, deformation stabilization monitoring, and damage pattern

observation. The test was terminated when the applied load dropped below 85 % of peak load, or the sudden excessive deformations rendering continued loading unsafe occurred.

3. Test results and analysis

3.1. Test observations and failure mode

Throughout the loading process, all specimens demonstrated a similar failure mode characterized by initial joint opening followed by progressive torsional deformation and concrete crushing on the bottom section. However, distinct specific damage patterns emerged depending on different prestress ratio n and bending-torsion ratio m . The failure modes of specimens are shown in Fig. 10. For descriptive clarity, the specimen's four main locations (east, south, west, and north sides) were

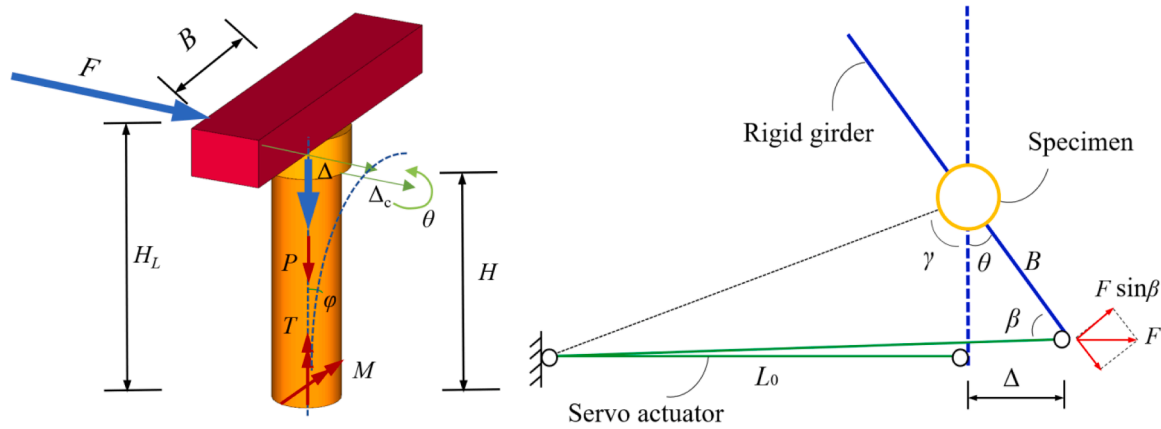


Fig. 7. Deformation of the specimen under compression-bending-torsion load.

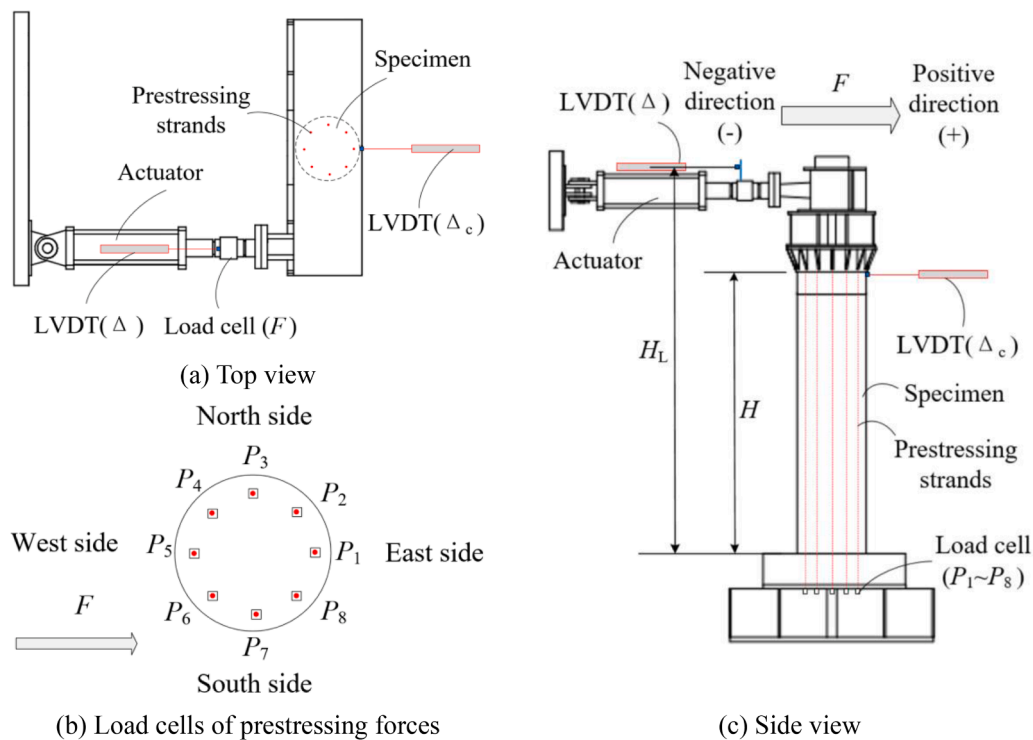


Fig. 8. Measurement arrangements of the forces and displacements.

labeled using the notation system defined in Fig. 8(b).

During initial loading stages, the tensile side of the joint opened, with opening widths increasing proportionally to horizontal displacements. The opened joint could be closed in the reverse loading, showing a certain self-centering performance. Then the diagonal cracks propagated upward from the bottom of the tower segment, after which the increment of joint opening diminished markedly. As cyclic loading progressed, the diagonal cracks developed continuously in both loading directions, and the cracks of different directions crossed at the bottom of the tower segment. Specimens with higher prestress ratios showed delayed joint opening, and produced longer and straighter cracks. The joints of specimens with lower prestress ratios opened earlier, and the cracks appeared earlier, concentrated in the bottom region of the segments, with wider width, longer length and more inclined angle. Specimens with lower bending-torsion ratio showed denser diagonal cracks, and the joint opening was less pronounced. With the continuous expansion of diagonal cracks, the concrete cover at the bottom of the segment gradually spalled, and the upper region of the segment tended

to slip along the widest diagonal crack on the side of the segment, coupled with torsional deformation of reinforcement bars within the principal crack.

When the applied load became large, intersecting diagonal and vertical cracks between the north side and compression side (east or west) induced concrete splitting tendencies, culminating in abrupt concrete crushing with localized spalling. Severe torsional distortion occurred on reinforcement bars at the north side of the segment, accompanied by partial fracture of bars, leading to rapid load degradation. Specimens with higher prestress ratios exhibited minimal torsional deformation but demonstrated sudden concrete crushing near the outer fiber of the compression side, showing expanded crushing regions coupled with pronounced reinforcement bars buckling rather than twisting. Conversely, specimens with lower prestress ratios displayed marked joint opening and significant torsional distortion at the bottom, accompanied by progressive concrete crushing initiating from north sides. On the other hand, specimens with higher bending-torsion ratio showed less torsional effect, and the crushing regions were closer

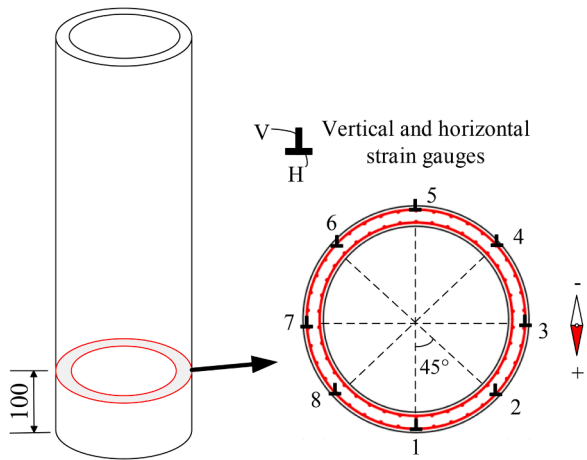


Fig. 9. Arrangements of strain gauges.

to the outer fiber of the compression zone (east or west), showing a more similar failure mode with compression-bending failure. Specimens with lower bending-torsion ratio developed denser diagonal cracks and exhibited severe reinforcement twisting and fracture at the north side of the segment. During the loading process, no obvious shear slippage was found in the horizontal joints.

3.2. Hysteresis curves

Based on the geometric relationships shown in Fig. 7, the torsional rotation angle and displacement angle of the specimen were calculated using horizontal displacements Δ and Δ_c . During testing, Δ and Δ_c increased proportionally. According to Eqs. (3) and (4), both torque and bending moment were proportional to horizontal force F , and using horizontal force to represent loads can effectively demonstrate the tower's resistance to loads with different bending-torsion ratios. Consequently, the F - Δ curves comprehensively reflect the relationships between bending moment/torque and their corresponding deformations. The obtained F - Δ hysteretic curves are shown in Fig. 11.

In early loading stages, all specimens exhibited small hysteretic loop areas with nearly linear loading and unloading behavior, indicating elastic response where joints remained closed, and no crack appeared yet. As joints opened, significant stiffness degradation occurred simultaneously with diagonal crack initiation at the compression-side of the bottom of the segment. Specimens with higher prestress ratios demonstrated delayed joint opening and correspondingly postponed stiffness degradations. The phases after stiffness degradation showed markedly increased hysteretic loop areas and residual displacements, with these effects being more pronounced in specimens with lower prestress ratios. Specimens with high prestress ratios exhibited smaller post-degradation loop areas, minimal residual displacements, and nearly overlapping loading and unloading paths, followed by rapid load deterioration after reaching peak load. During reversed loading, hysteretic loops maintained symmetrical patterns, opened joints fully closed upon unloading and reopened in the opposite direction under reverse loading. Compared to typical bow-shaped or spindle-shaped hysteresis curves of cast-in-place structures, prestressed concrete tower specimens under combined compression-bending-torsion loading displayed smaller enclosed loop areas (particularly in early loading cycles) and more evident pinching effects.

The experimental observations combined with the hysteretic curve characteristics demonstrate that under compression-bending-torsion cyclic loading, the joints exhibited repetitive opening and closing behavior with limited displacement magnitudes. This resulted in minimal plastic deformations during the initial loading stages, allowing the specimens to nearly recover their original state after each unloading

cycle, thereby demonstrating self-centering characteristics. However, in later loading phases, the joint opening does not increase significantly, and the deformation at the bottom of the segment began to increase significantly under the combined action of bending and torsion, resulting in substantial energy dissipation. Comparative analysis of specimens with varying prestress ratios and bending-torsion ratios further reveals that specimens with lower prestress ratios developed larger hysteretic loop areas in later stages, exhibiting progressively enhanced energy dissipation per cycle, while simultaneously achieving improved deformation ductility. Conversely, specimens with higher bending-torsion ratio displayed reduced hysteresis loop areas, with their mechanical behavior approaching compression-bending loading pattern, which better preserved the self-centering performance.

3.3. Backbone curves

The backbone curves were constructed by connecting the peak displacement points and their corresponding loads at each loading cycle, as shown in Fig. 12. The initial stiffness stages of the backbone curves for all specimens showed consistent tendency, indicating that within the tested parameter ranges, neither the prestress ratio nor bending-torsion ratio could exert significant influence on initial stiffness. This suggests that the initial stiffness under compression-bending-torsion load was predominantly governed by the tower segment's geometric dimensions. Specimens with higher prestress ratios demonstrated delayed transitions to the second stiffness phase and greater second stiffness magnitudes, attributable to the fact that elevated prestressing forces not only suppressed joint opening but also enhanced torsional resistance. However, this improvement came at the cost of reduced deformation capacity and ductility, resulting in significantly shortened ultimate displacements. For specimens with different bending-torsion ratios, the second stiffness magnitudes showed minimal differences under high prestressing forces. In contrast, under low prestressing force, specimens exhibited increased difference of the second stiffness, implying that torsional effects could adversely influence mechanical behavior, though such adverse effects could be mitigated by higher prestressing forces. Overall, the backbone curves maintained similar S-shaped across all specimens, yet substantial variations in ultimate capacity and ductility were observed.

3.3.1. Capacity index

Key mechanical characteristic states were extracted from the backbone curves and summarized in Table 3. The yielding state was determined according to the [43]. The ultimate state corresponded to the maximum load-bearing capacities, while the failure state was the point when the load dropped to 85 % of the peak load. For specimens exhibiting abrupt load drops, the failure states were assigned to the last stable cycle prior to concrete crushing. To quantitatively compare ultimate capacities (F_u) of all specimens, a capacity index (CI) was introduced:

$$CI = \frac{F_u}{F_{ub}} \quad (7)$$

Where F_{ub} is the ultimate capacity of the benchmark specimen CBT-800-0.22.

The CI values are listed in Table 3 and comparative analysis of average CI values are presented in Fig. 13. By comparing the results in Table 3 with results in [4], it can be found that the combined bending-torsion capacities of prestressed concrete towers are significantly lower than the bending capacity, indicating the significantly adverse effects of the torsion. Comparative analysis revealed progressive enhancement of combined capacities with increasing prestress ratios. Notably, specimen with the highest prestress ratio CBT-800-0.44 achieved a 122.04 % increase in load capacity compared to specimen CBT-800-0.11, significantly exceeding the CI improvement 84.56 % observed in the compression-bending test [4]. This result confirms that higher prestress ratios provide greater benefits for combined compression-bending-torsion capacity compare. This improvement can

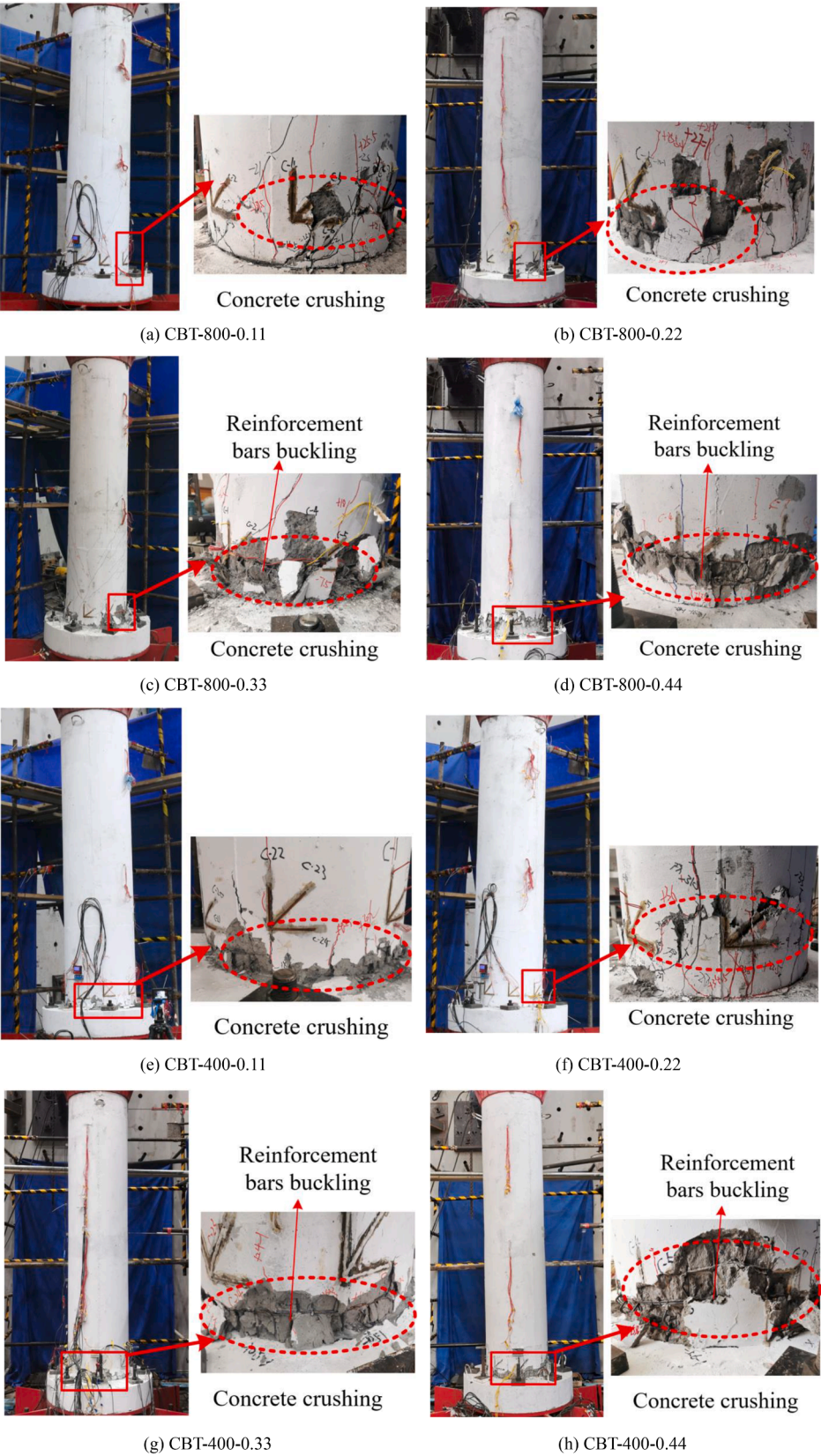


Fig. 10. Failure modes.

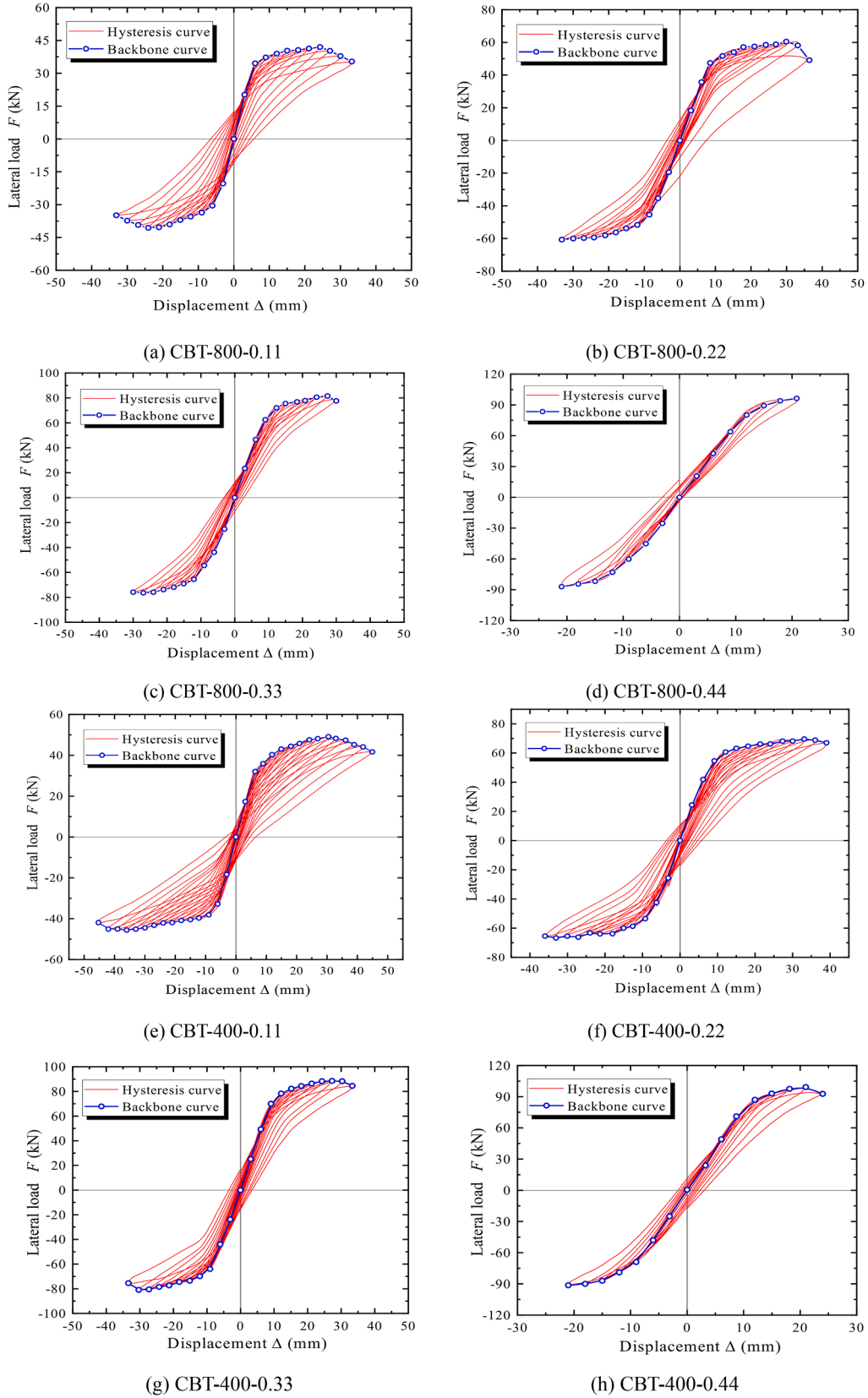


Fig. 11. Hysteresis curves of specimens.

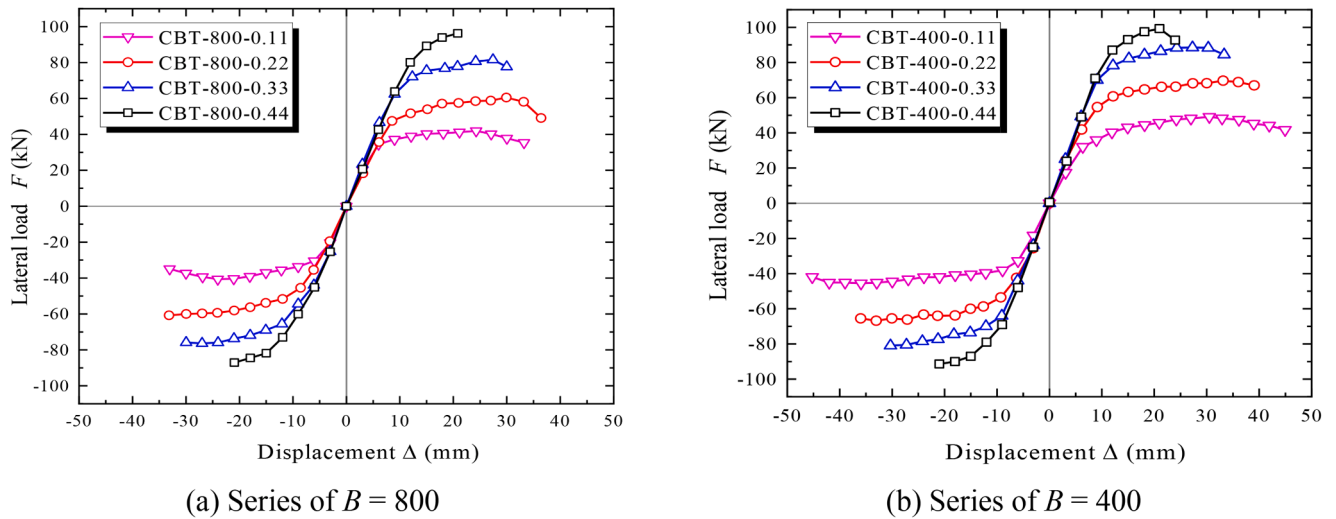


Fig. 12. Backbone curves of specimens.

Table 3

The characteristic states of backbone curves.

Specimen label	Loading direction	yield state		Ultimate state		Failure state		CI	DI
		F (kN)	Δ (mm)	F (kN)	Δ (mm)	F (kN)	Δ (mm)		
CBT-800-0.11	Positive+	35.95	7.61	41.95	24.28	35.66	32.86	0.69	4.32
	Negative-	32.61	8.02	40.62	24.04	34.91	33.06	0.67	4.12
	Average	34.28	7.82	41.29	24.16	35.28	32.96	0.68	4.22
CBT-800-0.22	Positive+	52.10	12.56	60.49	29.92	51.42	35.59	1.00	2.83
	Negative-	51.97	12.35	60.71	33.17	51.61	34.83	1.00	2.82
	Average	52.03	12.45	60.60	31.54	51.51	35.21	1.00	2.83
CBT-800-0.33	Positive+	72.07	12.60	81.54	27.43	77.60	29.99	1.35	2.38
	Negative-	66.70	13.04	76.41	26.98	75.87	29.98	1.26	2.30
	Average	69.39	12.82	78.97	27.21	76.73	29.99	1.30	2.34
CBT-800-0.44	Positive+	90.41	15.73	96.27	20.84	81.83	21.79	1.59	1.39
	Negative-	82.35	15.63	87.08	20.94	87.08	20.94	1.43	1.34
	Average	86.38	15.68	91.67	20.89	84.45	21.36	1.51	1.36
CBT-400-0.11	Positive+	41.23	8.09	49.12	30.49	41.75	44.83	0.81	5.54
	Negative-	38.16	8.99	45.57	35.94	41.94	45.25	0.75	5.03
	Average	39.70	8.54	47.34	33.22	41.85	45.04	0.78	5.27
CBT-400-0.22	Positive+	61.83	12.25	69.62	33.13	67.00	39.06	1.15	3.19
	Negative-	57.67	11.79	66.79	33.02	56.77	36.47	1.10	3.09
	Average	59.75	12.02	68.20	33.08	61.88	37.76	1.13	3.14
CBT-400-0.33	Positive+	79.07	12.73	88.50	27.28	84.50	33.33	1.46	2.62
	Negative-	70.59	12.68	80.95	30.33	75.50	33.40	1.33	2.63
	Average	74.83	12.71	84.72	28.81	80.00	33.36	1.40	2.63
CBT-400-0.44	Positive+	93.31	15.20	99.21	20.99	92.70	23.99	1.64	1.58
	Negative-	82.40	13.28	91.35	21.01	91.35	21.01	1.50	1.58
	Average	87.86	14.24	95.28	21.00	92.03	22.50	1.57	1.58

be attributed to the prestressing effects, which not only suppressed joint opening and enhanced bending capacity but also significantly increased the torsional resistance of the tower segments. Comparative analysis of specimens with different bending-torsion ratios revealed that ultimate capacities increased with decreasing bending-torsion ratio. Specifically, specimen CBT-400-0.11 exhibited a 14.66 % higher capacity than CBT-800-0.11. These results indicate that higher bending-torsion ratio mitigated torsional effects, enabling the specimens to sustain greater load. Furthermore, under higher prestress ratios, the capacity reduction caused by decreasing bending-torsion ratio was relatively smaller, reaffirming that increased prestressing forces can partially counteract the detrimental effects of torque.

3.3.2. Ductility index

The following equation is adopted to describe the ductility of the specimens:

$$DI = \frac{\Delta_y}{\Delta_u} \quad (8)$$

Where, Δ_y is the displacement of yielding state, Δ_u is the displacement of ultimate state.

The ductility indexes of all specimens are shown in Table 3 and Fig. 14. The results demonstrate that specimen CBT-400-0.11 exhibited a 25.06 % enhancement in ductility compared to CBT-800-0.11, indicating specimens with higher bending-torsion ratio achieved superior ductility. This phenomenon can be attributed to that the torque restricted the capacity of bending-induced joint opening, and the shear-torsion deformation capacity of concrete was limited. Thus, the torsional effects significantly reduced the ductility of specimens. When the bending-torsion ratio remained constant, ductility was significantly compromised by higher prestress ratios. Specimen CBT-800-0.44 exhibited a ductility coefficient of merely 1.36, while CBT-800-0.11 demonstrated a 209.51 % improvement. Similarly, CBT-400-0.11

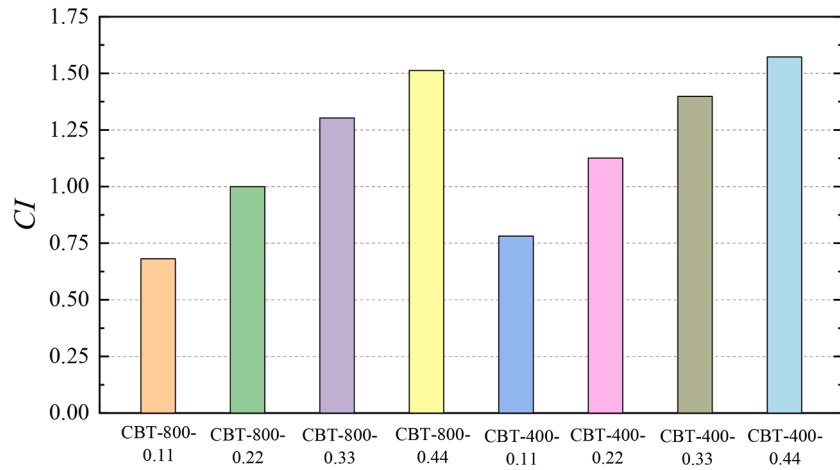


Fig. 13. Comparison of load-bearing capacity indexes of specimens.

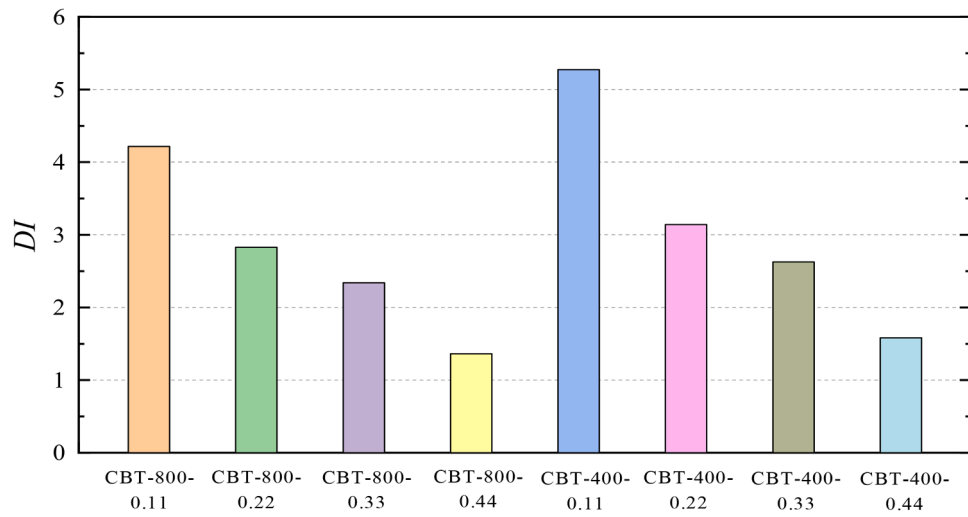
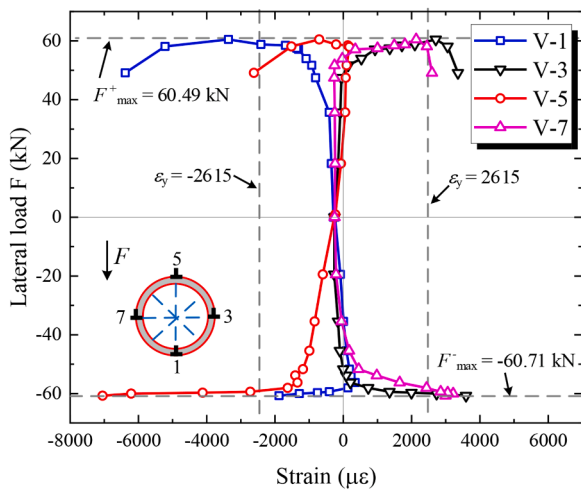
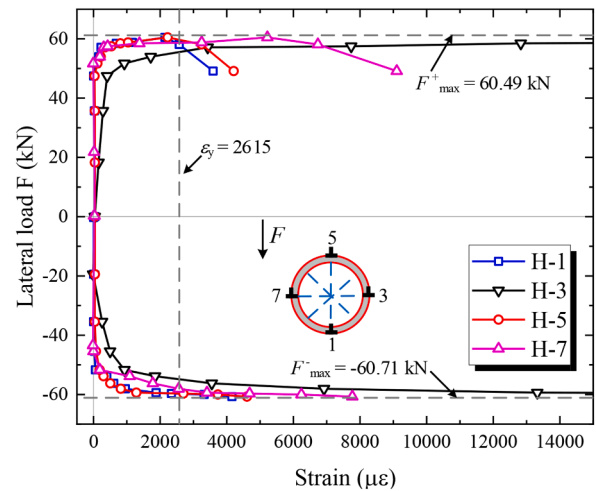


Fig. 14. Comparison of ductility indexes of specimens.



(a) Strains of vertical reinforcement bars



(b) Strains of horizontal reinforcement bars

Fig. 15. Load-strain curves of specimen CBT-800-0.22.

showed a 233.76 % ductility increase over CBT-400-0.44. Large prestressing forces not only suppressed joint opening but also restricted torsional deformation of the reinforcement bars. Reducing prestressing force effectively improved ductility, with greater enhancements when bending-torsion ratio is low.

3.4. Strain response

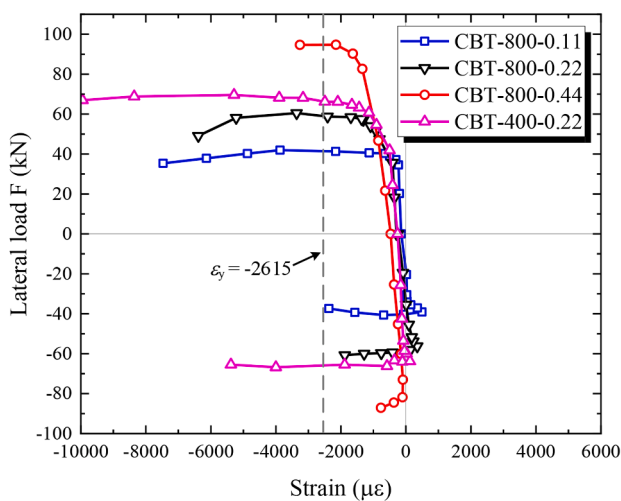
Strain data of reinforcement bars of tower segment at peak displacement points of each loading cycle were selected and connected to form the strain-load relationship. Taking CBT-800-0.22 as an instance, the strain-load curves of different locations are shown in Fig. 15, where positive values denote tensile strains, and negative values denote compressive strains. The labels in Fig. 9 are used to describe the corresponding reinforcement bars.

Vertical rebars V-1 and V-5 exhibited progressively increasing compressive strains during positive loading and negative loading, respectively. Both reached yielding strain before the peak load was reached. This behavior indicates that V-1 and V-5, positioned at the compression edge of the bending moment, primarily resisted flexural compressive stresses with minimal torsional contribution. During reverse loading, joint opening prevented V-1 and V-5 from effectively carrying tensile stresses, resulting in negligible tensile strains in the early loading stage. In the late loading stage, the vertical rebars entered the compressive plastic phase, thus the irreversible deformations caused them to remain in compression even throughout reverse loading. Conversely, vertical rebars V-3 and V-7 sustained tensile strains in both loading directions, with strains progressively increasing to yielding strain. Reinforcement bars V-3 and V-7 remote from the bending compression edge, predominantly resisted tensile stresses induced by torsion. Analysis of horizontal reinforcement strains (Fig. 15(b)) further corroborates this mechanism. Horizontal reinforcement H-3 and H-7 at north and south sides exhibited significantly larger tensile strains, reaching yield before peak load. This shows that the torsional deformation is significant on the lateral regions (north and south sides) of the tower segment under the combined load. Notably, H-3 demonstrated excessive tensile strain development far exceeding yielding strain, aligning with observed localized torsional damage on the specimen's north side during testing.

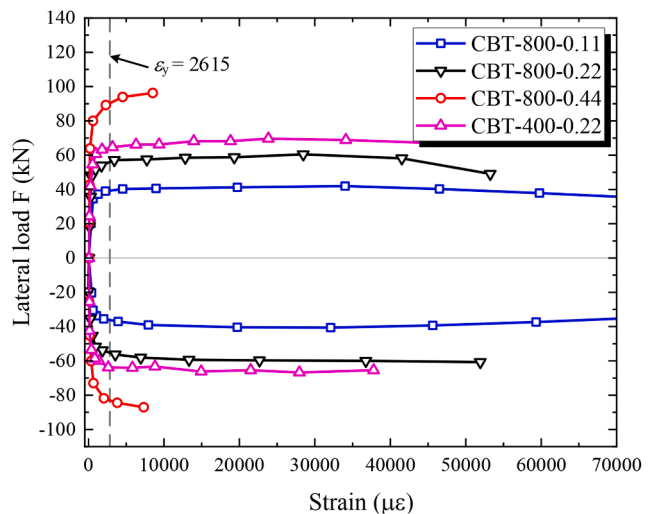
Fig. 16 compares strain characteristics across specimens using vertical strain V-1 and hoop strain H-3, which represent maximum deformation regions. As shown in Fig. 16(a), for specimens with identical bending-torsion ratio, decreasing prestress ratios resulted in lower

initial vertical strains but higher peak vertical strains. This phenomenon occurs because smaller prestressing forces induced more pronounced joint opening, reducing effective compression areas and thereby amplifying compressive strains at the bending compression edge. Notably, specimen with the largest prestress CBT-800-0.44 exhibited rapid load capacity attainment after reinforcement yielding followed by abrupt failure, demonstrating poor ductility. When the prestress ratio was constant, vertical strains increased significantly with increasing bending-torsion ratios, indicating enhanced bending dominance. Analysis of horizontal strains (Fig. 16(b)) revealed that increasing prestress ratios substantially reduced strains of horizontal reinforcements, demonstrating that prestressing force effectively enhances the resistance of torsional deformation. Furthermore, the strains of the horizontal reinforcements of the specimens with large bending-torsion ratio decreased significantly, highlighting the weakening of the torsional effect.

Fig. 17 illustrates the circumferential strain distribution of reinforcement at the positive ultimate state. Strain distribution of vertical reinforcements generally followed bending behavior patterns, with maximum compressive strains occurring at the bending compression edge (0° position in Fig. 17). Compressive strains at this location increased significantly with decreasing prestress ratios. Lateral regions of the tower segment exhibited tensile strains due to pronounced torsional effects. The compression zone range of vertical reinforcements expanded with higher prestress ratios, reaching nearly full-section compression in specimens with maximum prestressing, further validating that the prestressing forces effectively restrained torsional deformation and promoted bending-dominant behavior under combined loading. Conversely, decreasing the bending-torsion ratio substantially reduced both the compression zone range and strain magnitudes in vertical reinforcement bars. Horizontal reinforcements (Fig. 17(c)(d)) were in full-section tensile states, with tensile strains exceeding strains in vertical reinforcement bars. Lateral regions (particularly at location of 90°) showed peak tensile strains, demonstrating severe torsional deformation. Even at the bending compression edge (0° and 180°), the horizontal reinforcement bars retained tensile states due to their role in confining concrete under compression, though bending effects dominated at these regions. The tensile strains of horizontal reinforcement bars were increased with reduced both prestress ratios and bending-torsion ratios, corresponding to more significant torsional deformation.



(a) Vertical strain V-1



(b) Horizontal strain H-3

Fig. 16. Comparison of reinforcement strain between different specimens.

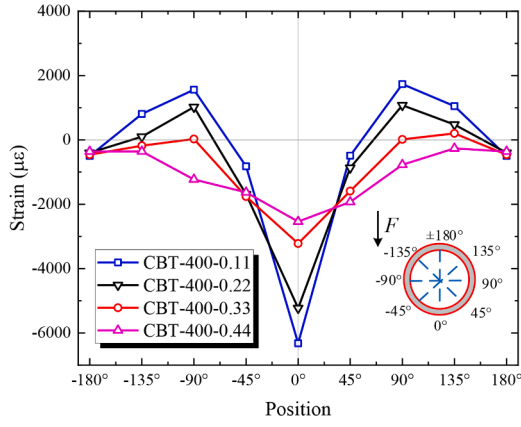
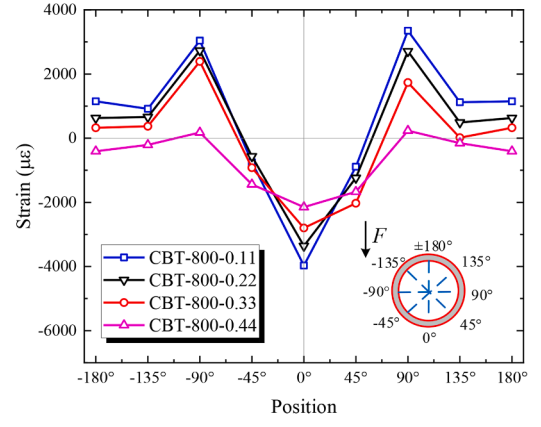
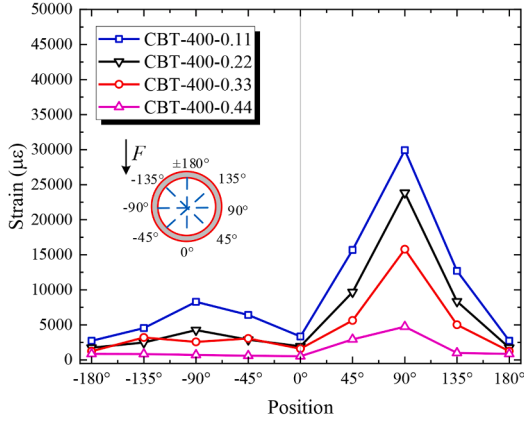
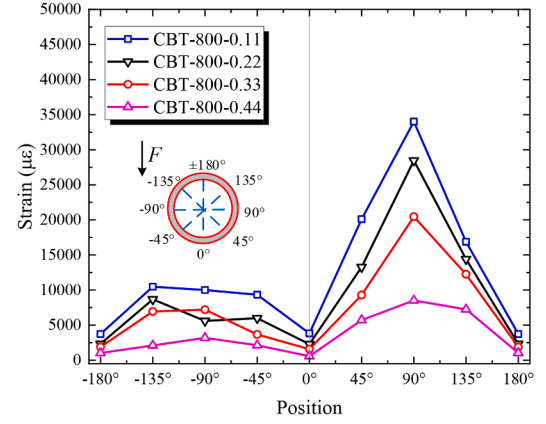
(a) Vertical strain distribution of specimens ($B = 400$)(b) Vertical strain distribution of specimens ($B = 800$)(c) Horizontal strain distribution of specimens ($B = 400$)(d) Horizontal strain distribution of specimens ($B = 800$)

Fig. 17. Strain distribution of specimens at ultimate state.

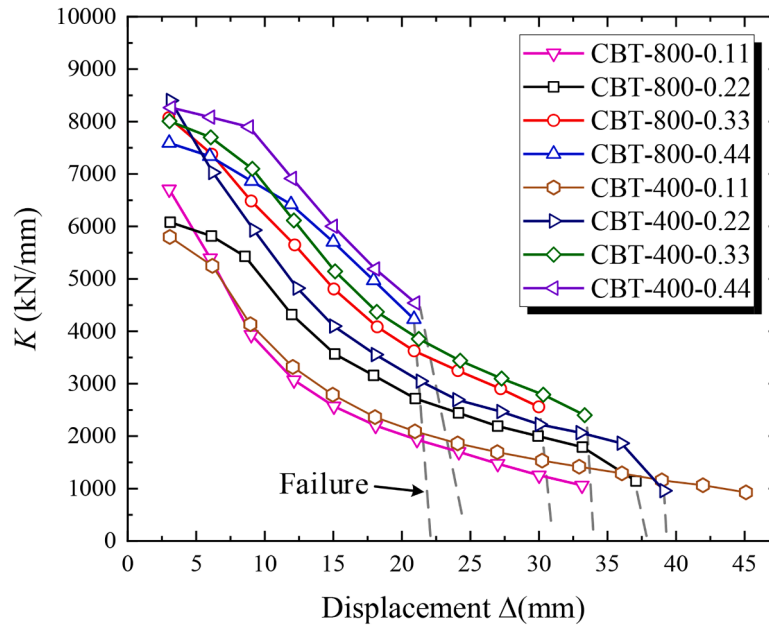


Fig. 18. Stiffness degradation curves of specimens.

3.5. Stiffness degradation

Analysis of the hysteretic curves reveals progressive degradation of secant stiffness in all specimens with increasing cycle numbers, demonstrating stiffness deterioration phenomena. To quantitatively evaluate stiffness degradation patterns across specimens, the secant stiffness K_i was introduced for assessment, with comparative results presented in Fig. 18. The secant stiffness K_i is defined as:

$$K_i = \frac{|+F_i| + |-F_i|}{|+\Delta_i| + |-\Delta_i|} \quad (9)$$

Where, $+F_i$ and $-F_i$ are the positive and negative peak loads in the i th loading cycle, respectively; $+\Delta_i$ and $-\Delta_i$ are the displacements corresponding to positive and negative peak loads in the i th loading cycle, respectively.

It can be found that the secant stiffness of specimens varied but their stiffness degradation trends exhibited consistent patterns. The degradation curves generally approximated power-law forms before failure state, with accelerated stiffness reduction during early loading stages. Correlated with experimental observations, this rapid initial degradation phase corresponded to limited crack development and negligible torsional deformations in the tower segments. The primary mechanism driving stiffness deterioration at this stage was attributed to joint opening-induced boundary constraint weakening, which significantly increased specimen deformations. As loading approached the yielding state, the joint opening was stabilized and the reduction of concrete compression areas was mitigated, while torsional deformations of the tower segment began to increase. This transition marked a decrease in stiffness degradation rates, with specimens exhibiting flatter degradation curves at lower prestress ratios. But after the ultimate displacement, the stiffness degraded sharply due to the abrupt failure of concrete crushing, as shown by the grey dashed line in Fig. 18. Notably, specimen with the highest prestress ratio CBT-800–0.44 displayed distinct degradation behavior: gradual stiffness reduction in early phases followed by accelerated degradation after yielding. This anomaly reflects weakened ductility under excessive prestressing forcing, where combined compression-bending-torsion loading led to rapid concrete crushing and premature stiffness degradation before ultimate state. In contrast, specimens with minimal prestress ratios CBT-800–0.11 and CBT-400–0.11, demonstrated lower secant stiffness magnitudes but slower degradation rates in the late loading stage. The load decreased slowly after ultimate state and showed better ductility. Comparative analysis of specimens with different bending-torsion ratios revealed that specimens with larger bending-torsion ratio showed higher initial stiffness and slower stiffness degradation, attributable to the reduced torsional effects and improved ductility.

3.6. Energy dissipation capacity

The enclosed area of hysteretic loops at each loading cycle characterizes the specimen's energy dissipation capacity. Larger loop areas indicate stronger energy dissipation capabilities. The equivalent viscous damping coefficient h_e is employed to quantify this energy dissipation capacity [44]:

$$h_e = \frac{1}{2\pi} \frac{S_{(ABD+CBD)}}{S_{(AFO+CEO)}} \quad (10)$$

Where, $S_{(ABC+CDA)}$ is the enclosed area of the hysteretic loop, representing the total energy dissipated during the corresponding loading cycle; $S_{(OBE+ODF)}$ is the sum of nominal elastic potential energies, calculated as the triangular areas shown in Fig. 19.

The evolution of equivalent viscous damping coefficients h_e with horizontal displacement Δ for all specimens is illustrated in Fig. 20. Overall, specimens under combined compression-bending-torsion loading exhibited superior energy dissipation capacity compared to

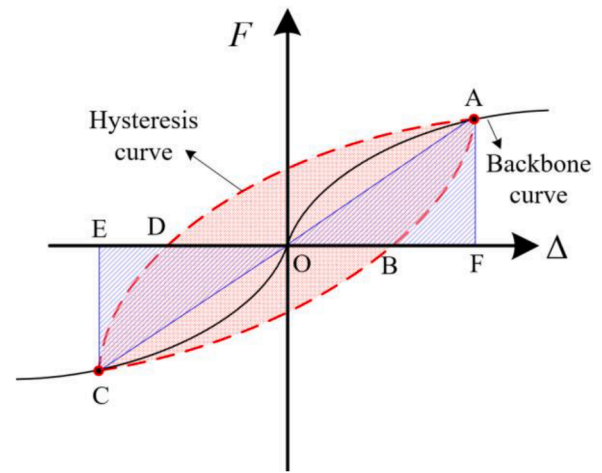


Fig. 19. Calculation of equivalent viscous damping factor.

specimens subjected to compression-bending load. During initial loading stages, h_e values remained relatively stable, gradually increasing with progressive displacement. In later phases, energy dissipation increased to different extents. Comparative analysis of specimens with different prestress ratios revealed that specimens with lower prestress ratios exhibited enhanced energy dissipation in later loading stages. The torsional resistance was reduced when the prestressing force was small, and the concrete diagonal cracked under torsion induced substantial plastic deformations during later loading phase, thereby generating progressive energy dissipation. Conversely, specimens with higher prestress ratios demonstrated suppressed torsional deformations in the early loading phase, limiting energy dissipation initially. However, as loading approached peak load, rapid torsional deformation growth coupled with excessive principal compressive strains under combined compression-bending-torsion loads, leading to abrupt concrete crushing and load degradation, which resulted in huge energy dissipation at ultimate displacements. For specimens with different bending-torsion ratios, h_e values were significantly lower at higher bending-torsion ratio. This trend indicates that increased bending-torsion ratio can diminish torsional effects, shifting mechanical behavior toward bending-dominance. The resultant improved self-centering characteristics in specimens with higher bending-torsion ratio reduced energy dissipation compared to torsion-dominated loads.

3.7. Residual displacement

Residual displacement Δ_r , defined as the unrecovered displacement at zero-load points of each hysteretic loop, serves as a clearer indicator of structural self-centering performance compared to energy dissipation metrics [45]. Fig. 21 presents the average (positive and negative) residual displacement values across specimens. During initial loading stages, Δ_r exhibited gradual and near-linear growth, primarily attributed to elastic deformations and reversible joint opening. However, the torque was increased with the progressive loading, inducing irreversible torsional deformation that accelerated the accumulation of residual displacement. Near the ultimate state, the residual displacement increased rapidly because of the huge plastic deformation of reinforcement bars, significantly weakening the self-centering performance. Specimens with lower prestress ratios exhibited earlier increase and larger ultimate magnitudes of residual displacement, attributed to the weakened torsional resistance of the tower segment under reduced prestressing forces, which accelerated and increased the torsional distortion. Conversely, specimens with higher prestress ratio maintained minimal residual displacement prior to ultimate state, but experienced rapid increase after ultimate state due to abrupt concrete crushing. Specimens with higher bending-torsion ratio showed superior

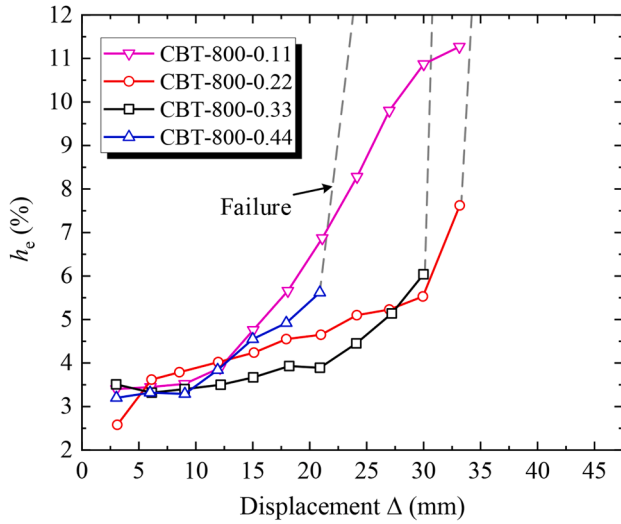
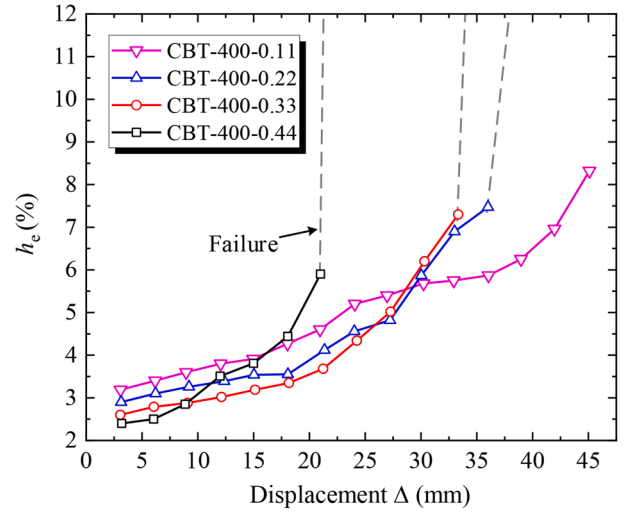
(a) Series of $B = 800$ (b) Series of $B = 400$

Fig. 20. Equivalent viscous damping coefficients of specimens.

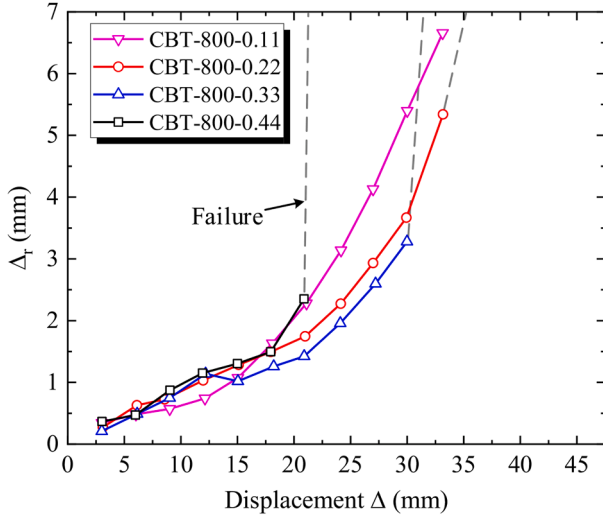
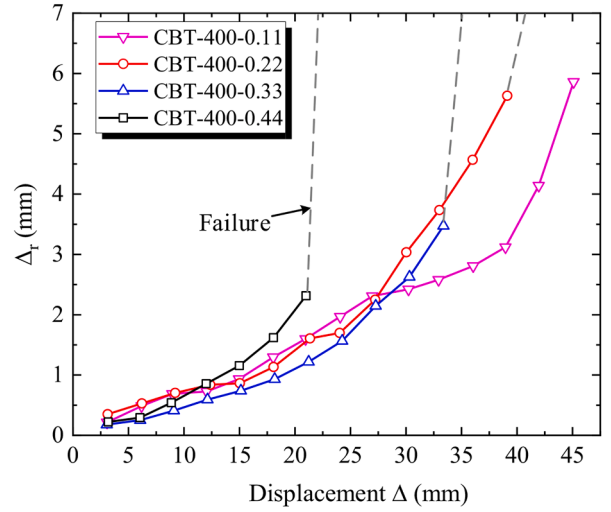
(a) Series of $B = 800$ (b) Series of $B = 400$

Fig. 21. Residual displacements of specimens.

early-stage self-centering performance due to their reduced torsional deformation, resulting in significantly smaller Δ_r during initial loading. However, progressive concrete crushing in later stages gradually increased residual displacements.

To more straightly compare residual displacement of specimens, the dimensionless quantity residual displacement ratio λ_r [46] was introduced as a direct indicator to describe the self-centering performance before the ultimate state. Considering that specimens with higher prestress ratio experienced abrupt after ultimate state, while specimens with lower prestress ratio sustained continued displacement growth beyond peak load, λ_r was uniformly calculated using ultimate displacement (Δ_u) and corresponding residual displacement ($\Delta_{u,r}$) for standardized comparison:

$$\lambda_r = \frac{\Delta_{u,r}}{\Delta_u} \quad (11)$$

The residual displacement ratios at ultimate states of specimens are shown in Fig. 22. For specimens with lower bending-torsion ratio, λ_r

decreased significantly with increasing prestress ratios. This indicates that prestressing forces substantially enhanced the torsional resistance of tower segments, thus the residual displacement before ultimate state was reduced, and the self-centering performance could be effectively improved. Conversely, specimens with higher bending-torsion ratio exhibited slight increases in residual displacement ratios with the increase of prestress ratio. The analysis presents that the residual displacement predominantly originated from bending-induced concrete crushing rather than torsion. Larger prestressing force increased compressive deformations in concrete before ultimate states, leading to increased residual displacement, which is more consistent with the residual displacement behavior under compression-bending load [4]. However, this increase remained slight due to the torsional effects to increase the residual displacement on specimens with low prestress ratios. Notably, specimen CBT-400-0.11 demonstrated a 6.49 % reduction in residual displacement ratio compared to specimen CBT-800-0.11. Despite larger failure residual displacements also existed in specimens with higher bending-torsion ratio, their smaller λ_r values suggest that

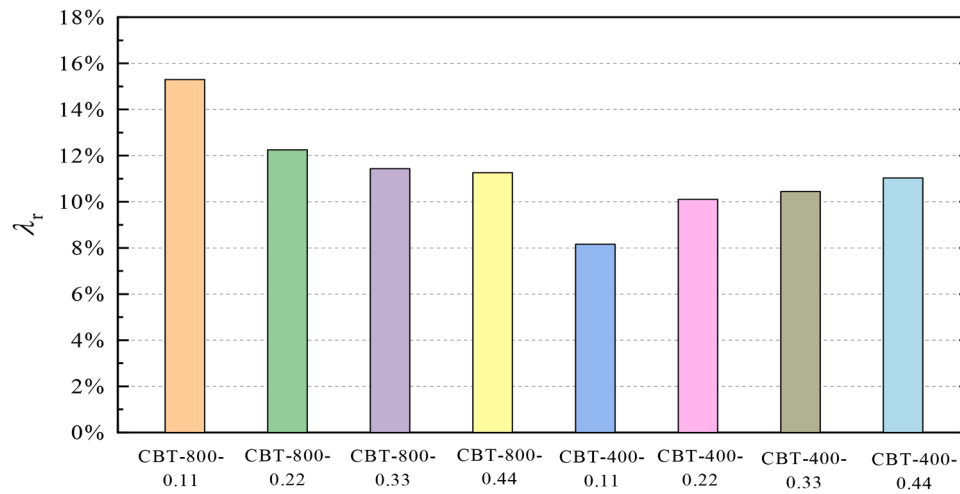


Fig. 22. Residual displacement ratios at ultimate state of specimens.

irreversible deformations primarily occurred after ultimate state, while effective self-centering performance was maintained prior to failure.

Comprehensive analysis of energy dissipation capacity and residual displacements reveals that prestressed concrete towers under combined compression-bending-torsion loading exhibit generally larger irreversible deformation due to torsional effects, which weakens the self-centering performance. In such case, higher bending-torsion ratios shift the mechanical behaviors toward bending-dominant modes, consequently reducing residual displacements and enhancing self-centering performance.

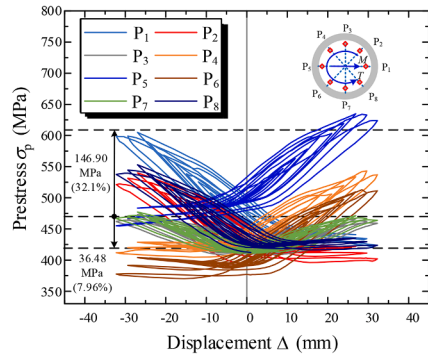
3.8. Prestress analysis

Prestress in the steel strands varied dynamically with tower segment deformations during loading. Fig. 24 illustrates the prestress σ_p variations of eight strands versus horizontal displacements. Strands P_1 , P_2 , and P_8 exhibited progressive prestress increases during negative loading with minor reductions in positive loading, while Strands P_4 , P_5 , and P_6 showed the inverse trend. This behavior occurs because horizontal displacement-induced joint opening under bending moments caused the rotation of tower segment, elongating strands on the tension side (increasing prestress) while shortening strands on the compression side (reducing prestress). Strands P_1 and P_5 , located at the tension edge of the bending moment, demonstrated significant prestress growth with the increase of displacement. Strands closer to the cross-sectional neutral axis exhibited minimal prestress fluctuations. Notably, P_3 and P_7 located at the centroidal axis experienced prestress accumulation under both loading directions, but this effect diminished with higher prestress ratios. In specimens with prestress ratio of 0.44, prestresses of P_3 and P_7 remained nearly constant, indicating neutral axis migration toward the centroidal axis under large prestressing forces, resulting in the larger compression zones. Most specimens exhibited abrupt prestress drops and large hysteretic loop areas in final loading cycles, attributable to localized concrete crushing near strands causing sudden shortening of strands. Specimens with higher bending-torsion ratio displayed more symmetric prestress distributions due to less torsional deformation and more bending-dominated behavior, resulting in uniform strand elongation and shortening patterns.

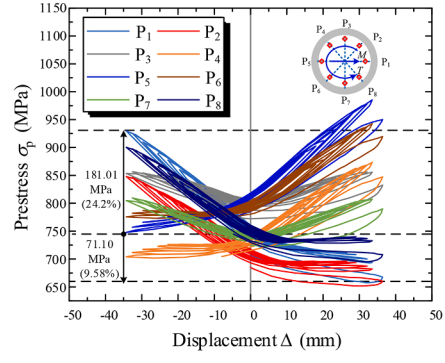
The prestress variations of Strand P_1 of all specimens at ultimate states are compared in Fig. 23. The analysis reveals that prestress increments increased significantly with the decrease of prestress ratios, as specimens with lower prestress ratios exhibited enhanced deformation capacity and greater strand elongation consequently. This trend was pronounced in specimens with higher bending-torsion ratios ($B = 400$). For instance, Strand P_1 in specimen CBT-400-0.11 showed a 54.3 %

prestress increase, whereas CBT-400-0.44 exhibited only a 5.42 % increase. Conversely, specimens with lower bending-torsion ratios ($B = 400$) demonstrated smaller prestress increment differentials: specimen CBT-800-0.11 achieved a 32.1 % increase, while specimen CBT-800-0.44 showed a 4.60 % increase. The substantial discrepancy in prestress increments existed between specimens with identical prestress ratio ($n = 0.11$) but different bending-torsion ratios. This is attributed to that the torsional effect is more pronounced in the specimens with lower bending-torsion ratio, thus the capacity of joint opening is constrained, indicating the adverse effect of torque on the bending deformation capacity of the tower under combined loading. However, specimens with the same prestress ratio 0.44 but different bending-torsion ratios exhibited minimal distinction of the prestress variation, proving again that the larger prestressing forces can more effectively restrict the torsional deformation. Additionally, higher prestress ratios reduced the disparity between prestress increments and decrements during cyclic loading, indicating again that the large prestress ratios make the neutral axis migrate toward the centroidal axis, leading to more uniform strand variations.

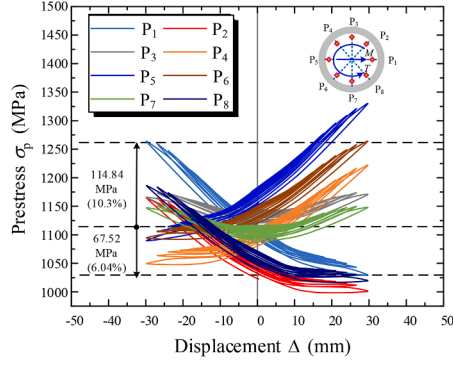
During cyclic loading, the specimens experienced prestress loss due to compressive deformations [47], resulting in a progressive reduction of prestress at zero horizontal displacement ($\Delta=0$) with increasing loading cycles, as can be seen from the intersections of prestress variation loops and x-axis in Fig. 24. The averaged prestress losses of all loading cycles are also shown in Fig. 24. Notably, specimens with $n = 0.11$ (CBT-800-0.11 and CBT-400-0.11) exhibited negative prestress losses during initial loading stages, gradually decreasing with the increase of horizontal displacement. Prestress loss did not exhibit significant growth until after the ultimate state. This behavior occurred because the tower exhibited helical elongation along inclined crack failure planes under torque before ultimate state, resulting in progressive elongation of prestressing strands during cyclic loading. The prestress losses only began to increase after the ultimate state, when the concrete gradually crushed. In contrast, all other specimens demonstrated monotonic prestress loss accumulation. At the same loading cycles, the prestress losses of specimens with higher prestress ratios were significantly greater. The prestress loss of specimen CBT-800-0.11 showed 124.49 % reduction compared to specimen CBT-800-0.44. This is attributed to the irreversible compressive deformations under larger prestressing forces that led to larger strand shortening. Specimens with lower prestress ratios showed smaller and more uniform prestress loss, reflecting superior ductility through less concrete crushing and irreversible deformations. On that other hand, comparative analysis of specimens with different bending-torsion ratios revealed that the specimens with higher bending-torsion ratio led to greater prestress losses



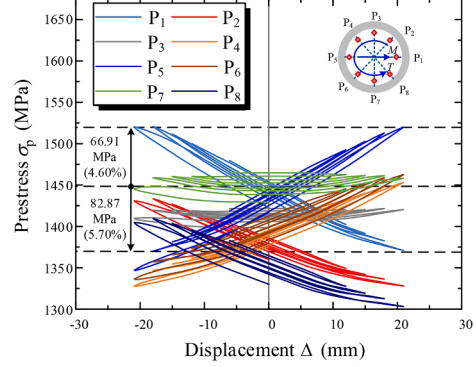
(a) CBT-800-0.11



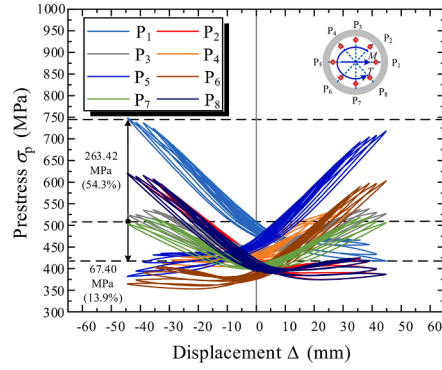
(b) CBT-800-0.22



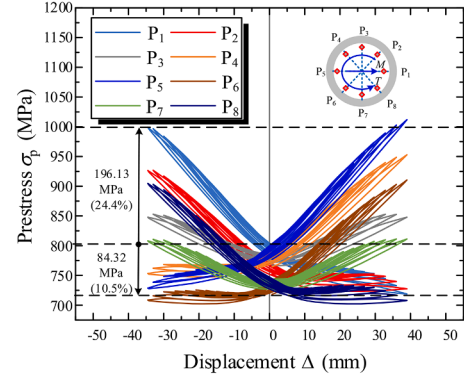
(c) CBT-800-0.33



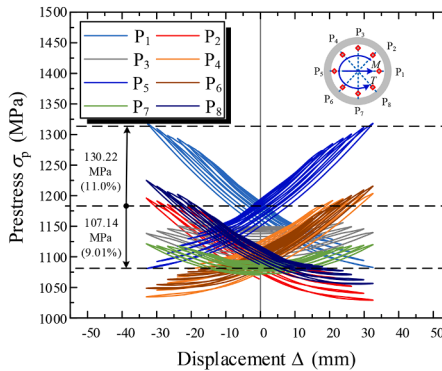
(d) CBT-800-0.44



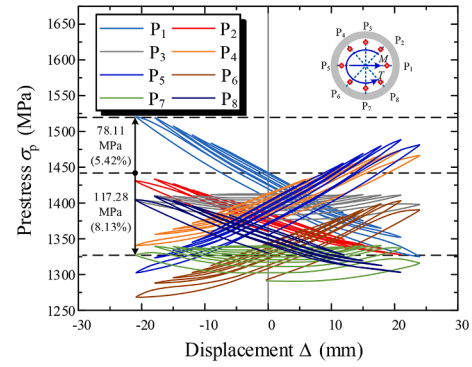
(e) CBT-400-0.11



(f) CBT-400-0.22



(g) CBT-400-0.33



(h) CBT-400-0.44

Fig. 23. Prestress variations of specimens.

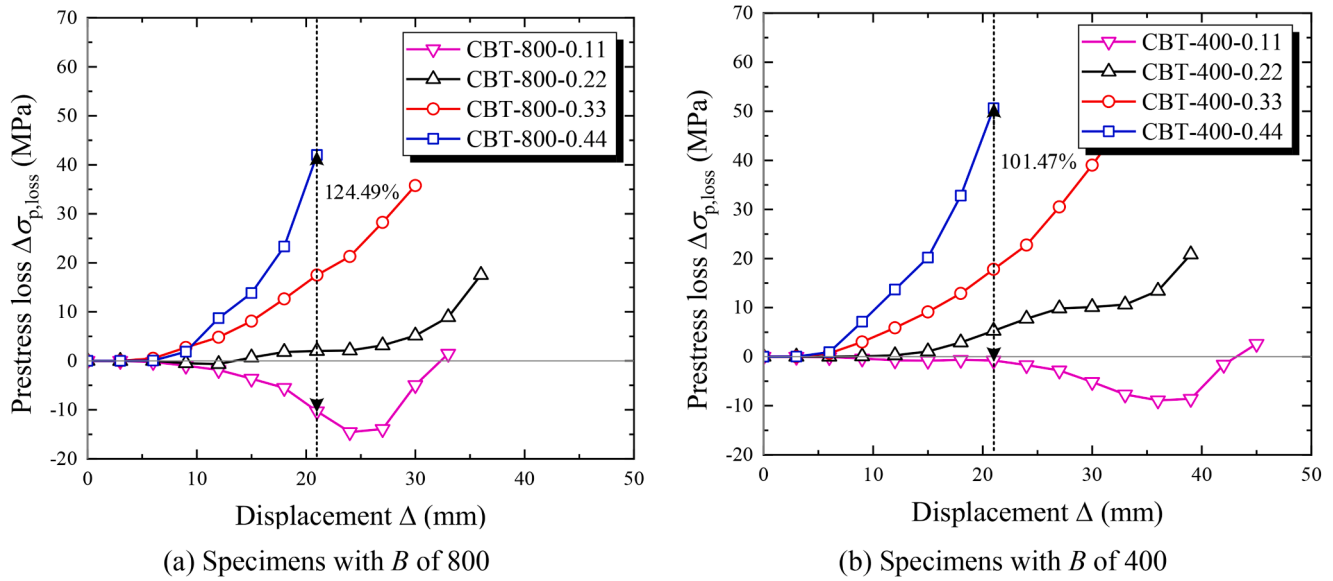


Fig. 24. Prestress loss of specimens.

due to the pronounced bending compressive deformation. However, the difference of prestress loss between specimens CBT-400-0.11 and CBT-400-0.44 was only 101.47 %, less pronounced than the 124.49 % difference between specimens CBT-800-0.11 and CBT-800-0.44. This contrast confirms that increased prestressing force can more effectively restrict the torsional deformation in specimens with lower bending-torsion ratio.

4. Evaluation of the ultimate capacity

Current design methods lack precise calculation formulas of ultimate capacity that account for the compression-bending-torsion behaviors of prestressed concrete towers. However, existing design methods can still be adapted to predict the ultimate capacity under combined loading. These conventional methods typically consider bending and torsional actions independently, neglecting their coupling effects. This oversimplification necessitates a critical evaluation of current verification

procedures of bending and torsional capacities. Therefore, the accuracy and applicability of current common design methods were evaluated based on the experimental results.

4.1. Ultimate bending capacity

Two primary calculation methods for the bending capacity of prestressed concrete towers are adopted in current design methods: the first relies on joint strength verification, while the second is governed by the ultimate bending capacity of the cross-section of tower segment. Both methods neglect torsional effects in their analytical frameworks, failing to account for the interaction between bending and torsion. Two methods are evaluated, respectively.

4.1.1. Bending strength of the horizontal joint

This method assumes that all horizontal joints remain unopened under bending moments, thereby ensuring strain distribution of joint

Table 4
Comparison of test and theoretical bending capacity of specimens under combined loads.

Specimen label	Loading direction	M_{ue} (kN)	M_{uc1} (kN)	M_{uc1} / M_{ue}	M_{uc2} (kN)	M_{uc2} / M_{ue}
CBT-800-0.11	Positive+	138.45	83.37	0.602	381.34	2.754
	Negative-	134.05	83.37	0.622	381.34	2.845
	Average	136.25	83.37	0.612	381.34	2.799
CBT-800-0.22	Positive+	199.62	122.14	0.612	366.69	1.837
	Negative-	200.36	122.14	0.610	366.69	1.830
	Average	199.99	122.14	0.611	366.69	1.834
CBT-800-0.33	Positive+	269.07	165.95	0.617	307.96	1.145
	Negative-	252.14	165.95	0.658	307.96	1.221
	Average	260.61	165.95	0.637	307.96	1.182
CBT-800-0.44	Positive+	317.69	198.83	0.626	234.41	0.738
	Negative-	287.36	198.83	0.692	234.41	0.816
	Average	302.52	198.83	0.657	234.41	0.775
CBT-400-0.11	Positive+	162.09	82.12	0.507	381.29	2.352
	Negative-	150.37	82.12	0.546	381.29	2.536
	Average	156.23	82.12	0.526	381.29	2.441
CBT-400-0.22	Positive+	229.75	123.27	0.537	365.77	1.592
	Negative-	220.39	123.27	0.559	365.77	1.660
	Average	225.07	123.27	0.548	365.77	1.625
CBT-400-0.33	Positive+	292.05	163.75	0.561	312.08	1.069
	Negative-	267.12	163.75	0.613	312.08	1.168
	Average	279.58	163.75	0.586	312.08	1.116
CBT-400-0.44	Positive+	327.40	196.99	0.602	239.53	0.732
	Negative-	301.47	196.99	0.653	239.53	0.795
	Average	314.44	196.99	0.626	239.53	0.762

interfaces adheres to the plane-section assumption. This assumption enables elastic stress verification for joint strength evaluation, requiring that tensile stresses on the tension side remain below the material's tensile strength and compressive stresses on the compression side do not exceed its compressive strength. Based on these assumptions, the ultimate bending capacity governed by joint strength is calculated as [13]:

$$M = \min \left\{ \left(f_t + \frac{P+N}{A_c} \right) \cdot W_c, \left(f_c - \frac{P+N}{A_c} \right) \cdot W_c \right\} \quad (12)$$

Where, f_t is the tensile strength of concrete; P and N are the prestressing force and vertical load, respectively; A_c and W_c are the section area and section modulus of the joint interfaces, respectively.

The bending capacity M_{uc1} governed by joint strength was calculated using and compared with experimentally measured moments M_{ue} , which were derived from lateral force F_{ue} by Eq. (12). From Table 4, it can be found that the mean value of M_{uc1}/M_{ue} is 0.601 with a standard deviation (SD) of 0.042, demonstrating systematic underestimation of the actual ultimate capacity by this method. This discrepancy arises from the core assumption of non-opening horizontal joints under bending, which contradicts experimental observations where joints could sustain increased bending moment and torque after opening. The results of specimens with lower prestress ratios will be more underestimated because the joint-opening capacity of these specimens are stronger, and the subjected bending moment can be subsequently increased after the joint opened. The calculated results of specimens with higher prestress ratios will be closer to the measured results due to the more stable joint connection. Specimens with lower bending-to-torsion ratio showed closer agreement between M_{uc1} and M_{ue} , not because the method can account for torsional effects, but due to torsion-induced reductions in measured bending capacity that coincidentally aligned with the underestimated predictions. Therefore, this simplified method can only be a conservative yet expedient preliminary design tool for conceptual stages, though its accuracy needs to be refined.

4.1.2. Bending capacity of the tower segment

This method presents the sectional bending capacity of tower segment based on the ultimate state of prestressed reinforced concrete circular sections, as adopted in Chinese design standards GB 50,011–2010 [48] and GB 50,135–2019 [49]. The method simplifies the trapezoidal stress distribution of compressive and tensile reinforcement bars into an equivalent rectangular stress block, assuming simultaneous yielding of both reinforcement bars and prestressing strands as the ultimate state. The resultant compression-bending capacity formula under this ultimate equilibrium condition is expressed as:

$$N \leq \alpha \alpha_1 f_c A_c - \sigma_{p0} A_p + \alpha f_{py} A_p - \alpha_t (f_{py} - \sigma_{p0}) A_p + (\alpha - \alpha_t) f_{sy} A_s \quad (13)$$

$$M \leq \left(\alpha_1 f_c A_c r_0 + r_0 f_{sy} A_s + r_p f_{py} A_p \right) \frac{\sin \alpha \pi}{\pi} + \left[(f_{py} - \sigma_{p0}) A_p r_p + r_0 f_{sy} A_s \right] \frac{\sin \alpha_t \pi}{\pi} \quad (14)$$

Where, A_c is the cross-section of tower segment; σ_{p0} is the decompression stress, which can be taken as the initial prestressing force when neglecting prestress losses; f_{py} is yielding strength of prestressing strands; r_0 is the middle radius of the tower segment; r_p is radius of positions of prestressing strands; α is the semi-angle coefficient of compression zone, which describes the compression area; α_t is the semi-angle coefficient of compression zone, which describes the compression area; α_t is derived as $\alpha_t = 1 - 1.5\alpha$, subject to a minimum value of 0.

The methodology accounts for the load-bearing contributions of vertical tensile reinforcement ($f_{sy} A_s$), tensile prestressing strands ($f_{py} A_p$), and compressive prestressing strands ($f_{py} A_p$), considering prestressed concrete towers as monolithic sections equivalent to cast-in-place structures. However, experimental strain and prestress analyses

revealed critical deviations from this assumption: discontinuities in vertical reinforcements at the horizontal joint prevented full utilization of tensile strength of reinforcement bars, while unbonded external prestressing strands exhibited negligible compressive resistance due to the absence of concrete confinement [49]. Consequently, both terms of tensile reinforcement bars and compressive strands are excluded, leading to the simplified formula:

$$N \leq \alpha \alpha_1 f_c A_c - \sigma_{p0} A_p - \alpha_t (f_{py} - \sigma_{p0}) A_p \quad (15)$$

$$M \leq \alpha_1 f_c A_c r_0 \frac{\sin \alpha \pi}{\pi} + (f_{py} - \sigma_{p0}) A_p r_p \frac{\sin \alpha_t \pi}{\pi} \quad (16)$$

By solving Eqs. (15) and (16), the bending capacity M_{uc2} governed by the tower segment's sectional resistance was calculated and compared with experimental results M_{ue} , as summarized in Table 4. The average value of M_{uc2}/M_{ue} is 1.567 (with SD of 0.042), which significantly overestimated the measured results. In fact, the results of specimens with lower prestress ratios were overestimated but the results of specimens with lower prestress ratios were underestimated.

This discrepancy arises because this method assumes full-section participation with simultaneous yielding of reinforcement bars and prestressing strands at the ultimate state, neglecting the effect of joint opening. However, according to the experimental observation, the cross-section of the tower segment was not fully stressed after the horizontal joint opened, while the prestress of the steel strands was difficult to reach the yielding strength. From the prestress variations from Section 3.9, it can be found that the smaller the prestress ratio of the specimen, the larger the bending moment increments required to achieve the assumed yielding strength of the steel strands. Conversely, specimens with higher prestress ratios experienced the capacity underestimation since it is assumed that the ultimate state is tension-controlled eccentric compression (yielding of steel strands) rather than the actual compression-controlled eccentric compression (concrete crushing). The torsional effects are not considered in this method, resulting in minimal computational variation across specimens with different bending-torsion ratios. These findings collectively demonstrate that this method produces non-conservative predictions within typical prestress ratio ranges, rendering them unsuitable for practical design applications under combined loading.

4.2. Ultimate torsional capacity

Similar to the bending capacity, the calculation method of torsional capacity is governed by two distinct analytical methods: joint-controlled mechanism and segment-controlled mechanism. The predictive accuracy and applicability of these methods were evaluated through experimental measure torsional capacities.

4.2.1. Torsional strength of the horizontal joint

This method assumes that the tower segment remains intact under torque, with failure mode governed by ultimate torsional strength of horizontal joints. It assumes that torsional resistance is primarily provided by the frictional force on the joint interface, thereby adopting a verification framework of torsional shear stress based on classical mechanical principles to calculate interfacial torsional strength. According to the Saint Venant's torsion theory of thin-walled sections (Fig. 25(a)) [14], the ultimate torque can be calculated as:

$$T = 2A_0 \cdot t \cdot \tau_t \quad (17)$$

With, $\tau_t = \frac{F_t}{t \cdot u_0} = \frac{\mu \cdot (P + N)}{t \cdot u_0}$

Where, A_0 denotes the area enclosed by the middle diameter of the tower wall; τ_t represents the torsional shear stress acting on the interface; F_t is the interface frictional force provided by prestressing forces and vertical loads; μ is the interface friction coefficient, taken as

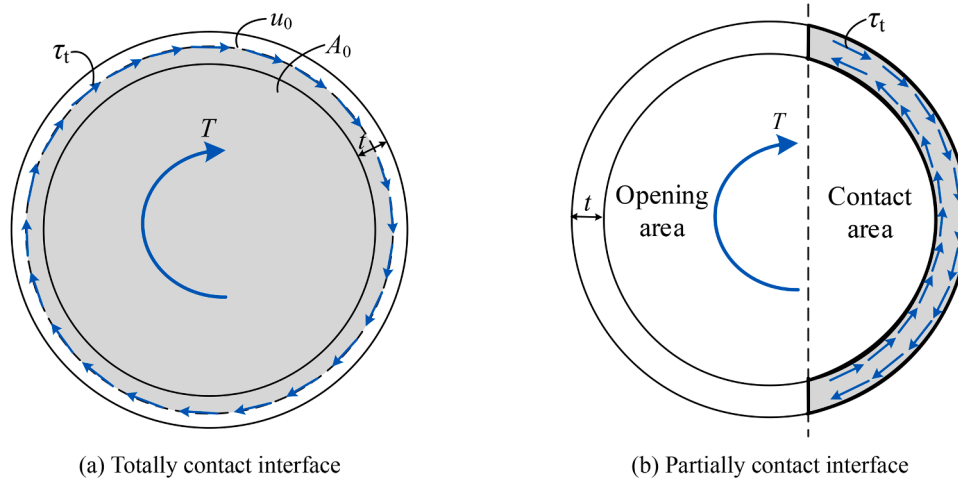


Fig. 25. Torsional shear stress mechanism of the horizontal joint.

0.7 for rough surfaces at precast concrete interfaces according to the specification in Eurocode 2 [50]; u_0 defines the perimeter corresponding to the middle diameter.

When horizontal joints open under bending moments, the effective contact area for frictional shear stress transfer is reduced, transforming the joint interface into a section with partially opening and partially contact. This geometric discontinuity invalidates the full-section shear flow model described by Eq. (17). For partially opened joints (Fig. 25 (b)), the ultimate torque of the joint is calculated as [36]:

$$T = \frac{\tau_t \cdot u_0 \cdot t^2}{3} = \frac{F_f \cdot t}{3} = \frac{\mu \cdot (P + N) \cdot t}{3} \quad (18)$$

For specimens under combined compression-bending-torsion loading in this study, horizontal joints generally opened prior to reaching the peak load. The joint-controlled torsional capacity T_{uc1} was calculated using Eq. (18) and compared with measured torsional capacity T_{ue} , as summarized in Table 5. The analysis revealed an average value of T_{uc1}/T_{ue} of 0.335 (with SD of 0.105), indicating severe underestimation of actual torsional capacity. This discrepancy exists because this method assumes that the tower fails immediately after the joint slips under torsion, which diverges from the actual failure mode that the

bending-torsion failure occurred at the bottom of tower segment after the joint opened. Experimental evidence demonstrates that the reinforcement bars and concrete of the tower segment substantially contribute to torsional resistance even after joint opening, with minimal slippage observed at the joint interface. Furthermore, although this method acknowledges that joint opening will create partial contact interfaces leading to capacity reduction, it over-simplifies the post-opening behavior by treating such interfaces as idealized open sections. Actually, the residual contact areas maintained structural integrity, enabling torsional resistance significantly exceeding that of idealized open sections, which is considered in the conservative assumption of this method. On the other hand, the horizontal joints were simply classified as either "open" or "closed", fails to quantitatively account for the effects of joint opening magnitudes and residual contact area on torsional resistance. This limitation precludes accurate prediction of post-opening torsional capacity. Consequently, the joint-controlled capacity calculation method exhibits excessive conservatism.

4.2.2. Torsional capacity of the tower segment

This method assumes that tower segments fail due to insufficient torsional capacity under combined compression-bending-shear-torsion

Table 5

Comparison of test and theoretical torsional capacity of specimens under combined loads.

Specimen label	Loading direction	T_{ue} (kN)	T_{uc1} (kN)	T_{uc1} / T_{ue}	T_{uc2} (kN)	T_{uc2} / T_{ue}
CBT-800-0.11	Positive+	33.56	6.20	0.185	103.16	3.074
	Negative-	32.50	6.20	0.191	103.16	3.174
	Average	33.03	6.20	0.188	103.16	3.123
CBT-800-0.22	Positive+	48.39	10.96	0.226	109.79	2.269
	Negative-	48.57	10.96	0.226	109.79	2.260
	Average	48.48	10.96	0.226	109.79	2.264
CBT-800-0.33	Positive+	65.23	16.35	0.251	117.27	1.798
	Negative-	61.13	16.35	0.267	117.27	1.919
	Average	63.18	16.35	0.259	117.27	1.856
CBT-800-0.44	Positive+	77.02	20.39	0.265	122.89	1.596
	Negative-	69.66	20.39	0.293	122.89	1.764
	Average	73.34	20.39	0.278	122.89	1.676
CBT-400-0.11	Positive+	19.65	6.04	0.308	102.94	5.240
	Negative-	18.23	6.04	0.331	102.94	5.648
	Average	18.94	6.04	0.319	102.94	5.436
CBT-400-0.22	Positive+	27.85	11.10	0.399	109.98	3.949
	Negative-	26.71	11.10	0.415	109.98	4.117
	Average	27.28	11.10	0.407	109.98	4.031
CBT-400-0.33	Positive+	35.40	16.08	0.454	116.90	3.302
	Negative-	32.38	16.08	0.496	116.90	3.610
	Average	33.89	16.08	0.474	116.90	3.449
CBT-400-0.44	Positive+	39.69	20.16	0.508	122.58	3.089
	Negative-	36.54	20.16	0.552	122.58	3.355
	Average	38.11	20.16	0.529	122.58	3.216

loading. Given the lack of torsional capacity formulas for circular sections under combined loading, the torsional capacity formula of rectangular section specified in GB 50,010–2010 [48] was adopted. By defining torsional yielding of reinforcement as the ultimate state, the method incorporates the synergistic torsional effects between concrete and reinforcement bars based on the variable-angle spatial truss theory, while accounting for the beneficial effects of prestressing force on enhancing concrete's torsional resistance. The derived torsional capacity formula is expressed as:

$$T_u = \beta_t \left(0.35f_t + 0.07 \frac{P+N}{A_c} \right) W_t + 1.2 \sqrt{\xi} f_{yh} \frac{A_{sh1} A_0}{s} \quad (19)$$

With, $\xi = \frac{f_{yv} A_{sv} s}{f_{yh} A_{sh1} u_0}$

Where, W_t represents the section plastic torsional modulus; ξ denotes the reinforcement strength ratio between vertical and horizontal reinforcements; A_{sv} is the total vertical reinforcement area; A_{sh1} refers to the cross-sectional area of one single horizontal reinforcement bar; s defines the spacing of horizontal reinforcement; f_{yv} and f_{yh} are the yield strengths of vertical and horizontal reinforcements, respectively; β_t is the reduction factor of concrete torsional capacity, governed by the member's shear-span ratio. For cases with large shear-span ratios where shear effects can be negligible (as in this study), β_t is set to 1.0.

The torsional capacity governed by the tower segment T_{uc2} was calculated through Eq. (19), and compared with experimental torque T_{ue} , as summarized in Table 5. The averaged value of T_{uc2}/T_{ue} is 3.132, indicating a general overestimation of the tower's ultimate torsional capacity. This discrepancy arises from the assumption of fixed-end boundary conditions at the bottom of tower segment, which neglects the adverse effects of horizontal joint opening. It can be found from the test results that the torsional capacity of the tower was significantly reduced due to the decreased contact area after joint opening. The overestimation magnitude of specimens with higher prestress ratios was reduced due to the torsional performance enhanced by the prestressing force, thus the value of T_{uc2}/T_{ue} is relatively smaller. For specimens with higher bending-torsion ratio, the actual torsional capacity was further reduced due to the increased joint opening, the results showed larger overestimation. Since the effects of joint opening cannot be considered in this method, the reliable predictions exist only when the joint opening is minimal, and the loading is close to pure torsion condition. Otherwise, the results are not conservative.

The above analysis reveals that existing design methods for bending and torsional capacities under combined compression-bending-torsion loading exhibit significant inaccuracies. The calculation method governed by joint strength systematically underestimates the actual ultimate capacity by assuming immediate failure upon joint deformation, while the calculation method controlled by tower segment capacity substantially overestimates the actual ultimate capacity through assumptions of idealized fixed boundary condition and monolithic section. In reality, the prestressed concrete towers under combined loading exhibited a hybrid failure mechanism: bending-torsion failure occurred at the bottom region of the tower segment accompanied by continuous joint opening. Thus, the ultimate state is governed by the composite effects considering both tower segment failure and joint opening. It is necessary to further analyze the ultimate capacity of prestressed concrete tower under combined loading while considering the actual combined mechanism, to propose a practical calculation method for the combined capacity.

5. Conclusions

The cyclic tests on eight prestressed concrete tower specimens under combined compression-bending-torsion loading were conducted to analyze the combined behavior, and the existing design methods were evaluated based on the test results. The following conclusions can be

drawn:

- (1) Under combined loading, the horizontal joint partially opened, and dense diagonal cracks propagated at the bottom of the tower. The concrete near the cracks crushed, accompanied by pronounced reinforcement twisting at the lateral region. Higher prestress ratios restrained joint opening and torsional deformation but resulted in abrupt concrete crushing. Larger bending-torsion ratios shifted failure modes toward bending dominance, with crushing localized near the edge of compression side.
- (2) The hysteresis loops displayed smaller enclosed loop areas and more evident pinching effects. Increasing the prestress ratio could significantly improve the ultimate combined capacity of specimens because the torsional deformation was restrained, but the ductility was significantly weakened. The larger bend-torsion ratio could reduce the influence of torque, thus enhancing both the ultimate capacity and ductility.
- (3) Torsional deformations weakened the self-centering performance of specimens, exhibiting larger energy dissipation and residual displacements compared to specimens under compression-bending load. Specimens with higher bending-torsion ratio demonstrated greater initial stiffness, slower stiffness degradation, and enhanced self-centering performance due to reduced torsional effects. At such case, increasing the prestress ratio would increase the residual displacement and weaken the self-centering performance. However, for specimens with low bend-torsion ratio, increasing the prestress ratio could decrease the energy dissipation and residual displacement instead, because of the enhanced resistance of torsional deformation.
- (4) The prestress increased with horizontal displacement, with the largest increments occurred at the tension edge of the bending moment and the smallest increments occurred near the neutral axis. Cyclic loading induced progressive prestress loss in strands. Specimens with lower prestress ratios exhibited enhanced deformation capacity, leading to greater prestress increments and reduced prestress loss. Lower bending-torsion ratios amplified torsional deformations, which reduced strand elongation and prestress increments while increasing the prestress loss.
- (5) The existing calculation methods of bending and torsional capacity were evaluated based on the test results. Existing methods cannot consider the combined bending-torsion effects and fail to capture the coupling behavior between tower segments and horizontal joints. The ultimate capacities were underestimated by the methods controlled by horizontal joints but overestimated by the methods governed by tower segments. Nevertheless, for towers with high safety requirements, the methods controlled by joints remain applicable as conservative yet expeditious preliminary tools for conceptual stages, though the accuracy needs to be refined.

It is recommended to conduct more studies to extend the range of the current work in this paper and further present the combined capacity calculation method of prestressed concrete towers, to guide practical engineering design. This will be investigated in-depth subsequently.

CRedit authorship contribution statement

Wei Ren: Writing – original draft, Methodology, Investigation, Formal analysis. **Xu-Hong Zhou:** Resources, Project administration, Conceptualization. **Ran Deng:** Visualization, Validation, Funding acquisition. **Wenyan Zeng:** Writing – review & editing, Validation. **Yu-Hang Wang:** Resources, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial

interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

No data was used for the research described in the article.

References

- [1] W. Ren, X.H. Zhou, J.S. Wang, et al., Pull-out behavior of the pre-installed lifting anchor with supplementary reinforcement for the precast concrete segments of wind turbine hybrid towers, *Eng. Struct.* 303 (2024) 117547.
- [2] J. Chen, J. Li, X. He, Ultimate torsional moment of dry horizontal joint for prefabricated concrete tower, *Struct. Des. Tall Spec. Build.* (2024) e2156.
- [3] Y.H. Wang, X.H. Zhou, L. Yang, et al., Current status and development trend of supporting structures for wind turbines, *Steel Construct.* 39 (10) (2024) 1–13.
- [4] W. Ren, X.H. Zhou, R. Deng, et al., Compression-bending behavior of thin-walled prestressed concrete tower with horizontal joint for wind turbines: experimental study and calculation model, *Thin-Walled Struct.* 211 (2025) 113154.
- [5] H. Ma, D. Zhang, Seismic response of a prestressed concrete wind turbine tower, *Int. J. Civil Eng.* 14 (2016) 561–571.
- [6] S.Z. Li, X.H. Zhou, Y.H. Wang, et al., Closed-form solution of fundamental frequency of steel-concrete hybrid wind turbine tower, *Int. J. Struct. Stab. Dynam.* 23 (03) (2023) 2350031.
- [7] S.Z. Li, X.H. Zhou, Y. Gao, et al., Damage detection of concrete segment in steel-concrete hybrid wind turbine tower, *Structures* (56) (2023) 105009.
- [8] J. Lana, P. Júnior, C.A. Magalhães, et al., Behavior study of prestressed concrete wind-turbine tower in circular cross-section, *Eng. Struct.* 27 (2021) 111403.
- [9] H.W. Ma, R. Meng, Optimization design of prestressed concrete wind-turbine tower, *Sci. China-Technol. Sci.* 57 (2) (2014) 414–422.
- [10] Z. Peng, C. Shukang, G. Bin, Simulation and optimization design of tower structure of wind generating set, *Int. Conf. Adv. Comput. Theory Eng.* 4 (2010) 309–312.
- [11] J. Chen, J. Li, X. He, Design optimization of steel-concrete hybrid wind turbine tower based on improved genetic algorithm, *Struct. Des. Tall Special Build.* 29 (2020) e1741.
- [12] Z. Zhou, C. Chen, X. Shen, et al., Conceptual design of a prestressed precast UHPC-steel hybrid tower to support a 15 MW offshore wind turbine, *Eng. Struct.* 321 (2024) 118939.
- [13] X. Huang, B. Li, X. Zhou, et al., Geometric optimization analysis of Steel-Concrete hybrid wind turbine towers, *Structures* 35 (2022) 1125–1137.
- [14] J. Grünberg, *Concrete Structures for Wind Turbines*, Wilhelm Ernst & Sohn, Berlin, Germany, 2013.
- [15] Z. Li, B. Xu, Wang Jiang, et al., Experimental and two-scale numerical studies on the behavior of prestressed concrete-steel hybrid wind turbine tower models, *Eng. Struct.* 279 (2023) 115622.
- [16] H. Ma, D. Zhang, Z. Ma, et al., Scale model experimental of a prestressed concrete wind turbine tower, *Wind Struct.* 21 (3) (2015) 353–367.
- [17] A. Palermo, S. Pampanin, D. Marriotti, Design, modeling, and experimental response of seismic resistant bridge piers with posttensioned dissipating connections, *Journal of Struct. Eng.* 133 (11) (2007) 1648–1661.
- [18] Y.C. Ou, P.H. Wang, S. Tsai, et al., Large-scale experimental study of precast segmental unbonded posttensioned concrete bridge columns for seismic regions, *J. Struct. Eng.* 136 (3) (2009) 255–264.
- [19] Z.Y. Bu, Y.C. Ou, J.W. Song, et al., Cyclic loading test of unbonded and bonded posttensioned precast segmental bridge columns with circular section, *J. Bridge Eng.* 21 (2) (2016) 04015043.
- [20] L. Zhang, K. Feng, M. Li, et al., Analytical method regarding compression-bending capacity of segmental joints: theoretical model and verification, *Tunnell. Underground Space Technol.* 93 (2019) 103083.
- [21] J.M. Avanaki, M. Abedi, A. Hoseini, Experimental and numerical-based design of hybrid steel fiber reinforced concrete tunnels, *Magaz. Concrete Res.* 72 (14) (2019) 720–733.
- [22] M. Zhang, G. Zhang, W. Huang, et al., Study on mechanical performance of prestressed assembled towering wind tower, *J. Build. Struct.* 43 (05) (2022) 62–78. In Chinese.
- [23] J.K. Tan, W.L. Zheng, Y. Zhou, et al., Study on the compression-bending capacity of horizontal joints in prestressed concrete towers for wind turbines, *Structures* 72 (2025) 108292.
- [24] J.K. Tan, M.S. Wang, M.N. Su, et al., Combined compression and bending resistance of steel fibre reinforced concrete tower for wind turbine, *Thin-Walled Struct.* 209 (2025) 112940.
- [25] Y.H. Wang, S. Xu, X.H. Zhou, et al., Lateral load behavior of prestressed segmental concrete towers with hollow tapered geometry, *Thin-Walled Struct.* 209 (2025) 112914.
- [26] R. Deng, X.H. Zhou, Y.H. Wang, et al., Behaviour of tapered CFDST columns with large hollow ratio under combined loads, *J. Construct. Steel Res.* 196 (2022) 107388.
- [27] L.H. Han, G.H. Yao, Z. Tao, Behaviors of concrete-filled steel tubular members subjected to combined loading, *Thin-Walled Struct.* 45 (6) (2007) 600–619.
- [28] J. Nie, Y. Wang, J. Fan, Experimental research on concrete filled steel tube columns under combined compression-bending-torsion cyclic load, *Thin-Walled Struct.* 67 (2013) 1–14.
- [29] R. Deng, X.H. Zhou, Y.H. Wang, et al., Behaviour of tapered concrete-filled double-skin steel tubular columns with large hollow ratio under combined compression-bending-torsion loads: experiments, *J. Build. Eng.* 57 (2022) 104876.
- [30] C. Jiao, Y. Liu, B. Wei, J. Li, et al., Seismic performance of circular reinforced concrete columns subjected to compression, bending and torsion with low and moderate shear span ratio, *Arch. Civil Mech. Eng.* 21 (3) (2021) 135.
- [31] Y.M. Selmy, E.F. El-Salakawy, Behaviour of circular concrete bridge columns internally reinforced with GFRP under reversed-cyclic loading including torsion, *Structures* 59 (2024) 105680.
- [32] W. Ren, X.H. Zhou, Z.L. Fu, et al., Static behavior of large-diameter stiffened steel tubes for wind turbine towers under combined compression-bending-torsion load, *Thin-Walled Struct.* 175 (2022) 109272.
- [33] R. Deng, X.H. Zhou, W.D. Ji, et al., Coupled behaviour and strength prediction of tapered CFDST columns with large hollow ratios for wind turbine towers, *Eng. Struct.* 289 (2023) 116287.
- [34] Y.S. Liu, X.H. Zhou, Y.H. Wang, et al., Seismic behavior of prestressed concrete filled steel tubular lattice tower subjected to combined compression-bending-torsion, *J. Construct. Steel Res.* 204 (2023) 107883.
- [35] Y.H. Wang, W. Z. X.H. Zhou, et al., Performance of hollow ultra-high-performance concrete-high strength steel columns for wind turbine towers with different confinement forms under compression-bending load, *Thin-Walled Struct.* 208 (2025) 112749.
- [36] C. Von Der Haar, S. Marx, Design aspects of concrete towers for wind turbines, *J. South Afr. Inst. Civil Eng.* 57 (4) (2015) 30–37.
- [37] C. Kang, S. Hartwig, S. Marx, Behavior of externally prestressed segmental towers' dry joint under torsion effects, *Struct. Concrete* 20 (2019) 1350–1357.
- [38] J. Chen, W. Lin, J. Li, Ultimate torsional moment of dry horizontal joint for prefabricated concrete tower, *Struct. Des. Tall Spec. Build.* (2024) e2156.
- [39] F. Klein, F. Füllr, T. Betz, et al., Experimental study on the joint bearing behavior of segmented tower structures subjected to normal and bending shear loads, *Struct. Concrete* 23 (2022) 1370–1384.
- [40] F. Füllr, F. Klein, S. Hartwig, et al., Experimental and analytical study on the load-bearing capacity of segmented tower structures with dry joints under combined loading, *Struct. Concrete* (2024) 1–14.
- [41] GB/T 50081-2019, Standard For Test Methods of Concrete Physical and Mechanical Properties, Ministry of Housing and Urban-Rural Development of PRC, Beijing(China), 2019.
- [42] GB/T 228.1-2010, Metallic Materials-Tensile Testing-Part 1: Method of Test at Room Temperature, Standardization Administration of China, Beijing(China), 2010.
- [43] R. Park, Evaluation of ductility of structures and structural assemblages from laboratory testing, *Bullet. New Zealand Natl. Soc. Earthq. Eng.* 22 (3) (1989) 155–166.
- [44] JGJT 101-2015, Specification For Seismic Test of Buildings, China building industry press, Beijing(China), 2015.
- [45] K. Kawashima, G.A. MacRae, J.I. Hoshikuma, et al., Residual displacement response spectrum, *J. Struct. Eng.* 124 (5) (1998) 523–530.
- [46] J. RuizGarcía, E. Miranda, Residual displacement ratios for assessment of existing structures, *Earthq. Eng. Struct. Dyn.* 35 (3) (2006) 315–336.
- [47] Y.S. Liu, Y.H. Wang, X.H. Zhou, et al., Compression-bending behavior of prestressed concrete filled steel tubular tower for onshore wind turbines, *Thin-Walled Struct.* 209 (2025) 112921.
- [48] GB50010-2010, Code For Design of Concrete Structures, Ministry of Housing and Urban-Rural Development of PRC, Beijing(China), 2015.
- [49] GB50135-2019, Code For Design of High-Rising Structures, Ministry of Housing and Urban-Rural Development of PRC, Beijing(China), 2019.
- [50] EN 1992-1-1:2004, Eurocode 2: Design of Concrete structures, Part 1-1: General rules and Rules For Buildings, European Committee for Standardization, Brussels (Belgium), 2004.